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ABSTRACT

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1. Introduction

A rotating machine in steady operation does not produce arbitrary signals. Its vibration is the output of a mechanism with finitely many degrees of freedom, observed through a sensing chain that introduces its own bounded distortions, and the set of waveforms it can emit while healthy is correspondingly restricted. Load, speed, mounting, sensor gain and acquisition phase move the observed signal about, but they move it within a region rather than across an unbounded space.

This paper takes that restriction as its governing premise and states it sharply, because it is the premise that makes the rest of the construction either meaningful or vacuous: *physically generated signals do not exhibit unbounded or adversarial variation, and departure from a compact nominal region is what an anomaly is*. The claim is not that nominal signals are often well behaved. It is that the nominal set is relatively compact, and therefore has a finite extent that can be estimated from finitely many observations. If that fails, no finite boundary exists, and a method built on estimating one is not merely inaccurate but meaningless.

Stating the premise this way makes it refutable, and Section 9 is arranged so that it could have been refuted. A process whose nominal variation did not concentrate would show a boundary estimate that kept growing with the number of observations rather than settling; the experiments measure exactly that quantity, and report both the regime in which it settles and the regime in which it does not.

Diagnosis as inverse recovery. An observation is the output of a forward process. A vibration window x is produced by a mechanism in some health state θ , acted on by nuisance factors η that carry no diagnostic information, so that $x = F(\theta, \eta)$. Recovering θ from x is an inverse problem, and in the classical sense an ill-posed one: the forward map is not injective, because distinct (θ, η) pairs can produce the same observation, and its inverse is unstable where it exists.

The usual response is to identify the forward model and invert it. For rotating machinery this is neither tractable nor necessary. It is intractable because a faithful forward model of a bearing in an installed drivetrain requires contact mechanics, lubrication state, structural transmission paths and sensor dynamics that are not identifiable from routine monitoring data. It is unnecessary because diagnosis does not require θ itself. It requires θ only up to the nuisance factors: two windows differing by mounting phase or sensor gain must receive the same diagnostic verdict, so the object to be recovered is not a point but an equivalence class.

That observation is the central claim of this work, and we state it in the form used in the companion technical note [7]: diagnostic monitoring can be formulated as inverse recovery over a quotient space induced by nuisance transformations of the observation process.

The reformulation changes what must be estimated. In the quotient space $\mathcal{X}/G_{\text{nuis}}$, the nuisance orbit of a nominal signal collapses to a single element, the nominal set becomes a region rather than a scattered cloud, and the diagnostic question becomes membership in that region. Three things then follow that do not follow in the ambient space. Membership is decidable from nominal data alone, so no fault examples are needed. The region has a finite extent, because bounded nuisance makes the orbit relatively compact. And that extent is a scalar, which is exactly the kind of quantity for which distribution-free statistical guarantees exist.

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What the premise buys, quantitatively. The last point is where a qualitative geometric argument becomes an engineering statement. If the nominal region has a finite radius, then estimating that radius is a tolerance-limit problem, and the coverage of the estimate can be certified without assuming a distribution for the residual. The certification is not asymptotic and does not appeal to a limit: at a target coverage of 98%, forty-nine nominal observations suffice, and the attained coverage is known exactly rather than bounded.

That number is the operational form of the premise. A regime that can be bounded from tens of observations is compact in the sense the premise asserts; a process without that structure would require the estimate to keep growing. The same analysis also delimits its own validity, and this is worth stating in the introduction rather than burying in the results: below forty-nine observations no boundary is certifiable at all, and between forty-nine and ninety-eight the certified estimator is the sample maximum, so it can appear to have converged while remaining maximally variable. A commissioning length chosen just above the floor satisfies the guarantee and is nonetheless a poor choice.

Contributions.

1. A quotient-space formalization of anomaly detection, with two results that carry the argument: the nominal orbit is relatively compact under bounded nuisance (Proposition 1), and a correct diagnostic decision factors through the quotient (Proposition 2), which is what licenses seeking one shared representation rather than a per-asset classifier.
2. A distribution-free coverage guarantee for the locally estimated operational radius (Proposition 7), given with proof. The statistical machinery is classical; the contribution is recognizing that the per-asset diagnostic radius is exactly a distribution-free tolerance limit on the nominal residual. Two consequences follow that are specific to the diagnostic setting and, to our knowledge, have not been stated for it: a minimum commissioning length (Corollary 6), and a degenerate range above that floor in which the estimator is the sample maximum (Corollary 7).
3. A saturation criterion connecting this formulation to the functional–topological account of Di Santi [8], with an explicit statement of what rapid convergence means quantitatively (Definition 3, Proposition 4), together with an operator adequacy criterion showing that coverage alone cannot certify a representation: two operators calibrated identically, at the same attained coverage, differ by more than two orders of magnitude in what they accept.
4. Validation across three tiers of increasing realism: synthetic assets under controlled nuisance, CWRU under four operating regimes with an oracle-versus-online calibration ablation, and NASA IMS run-to-failure bearings commissioned independently and with no fault labels at any stage.

Scope. Two boundaries should be set at the outset. The framework addresses membership in a nominal region, not fault classification: it reports that an asset has left its region, not which failure mode it has entered, and the taxonomy in Section 6 grows from expert feedback rather than from the geometry alone. And the guarantee is conditional on exchangeability of the nominal commissioning residuals (Assumption 4), which is an assumption about the commissioning window and not a property of the method; Section 9 reports a test of it rather than presuming it holds.

2. Related Work

Condition monitoring and fixed thresholds. Industrial vibration monitoring is dominated by band-limited scalar indicators — root-mean-square velocity, kurtosis, crest factor, envelope band energies — compared against thresholds. The reference framework is ISO 20816 [12], which specifies evaluation zones in terms of broadband vibration magnitude and, importantly for what follows, distinguishes absolute criteria from criteria based on change relative to a baseline established for the individual machine. The standard is explicit that zone boundaries are indicative and that a machine’s own history is the more reliable reference.

That distinction is the practical origin of the present work. Absolute thresholds transfer poorly across assets because the same health state produces different indicator values under different mounting, load and sensor placement, and the resulting practice — retune the threshold per machine — is effective but unprincipled: it has no statement of what coverage the tuned threshold attains, and no rule for how many observations are needed before it may be trusted. Sections 7 and 9 supply both for the quantity used here.

Novelty detection and reconstruction-based methods. Detecting departure from a nominal set without fault examples is the subject of novelty detection [2] and one-class methods, of which the support-vector formulation of Schölkopf et al. [20, 21] is the canonical instance: it estimates a region containing a prescribed fraction of the nominal mass, which is a level-set estimation problem in the sense of Devroye and Wise [6] and Scott and Nowak [22]. Reconstruction-based detection replaces the explicit region with an operator trained to reproduce nominal data, using the residual as the score [11, 18].

This paper is reconstruction-based, and takes the level-set reading seriously rather than as an analogy: the decision rule is membership in an estimated region, and Section 5 treats the radius as an estimate of that region’s extent, with the fill-distance argument of Proposition 6 controlling how the estimate approaches it. The difference from the one-class literature is not the detector but what is claimed about it. A one-class boundary is usually reported through its empirical false-alarm rate; here the boundary carries a distribution-free coverage statement with an exactly known attained level, and a minimum sample size below which no such statement is available.

Learning as an inverse problem. Treating estimation from examples as an ill-posed inverse problem requiring regularization goes back to Hadamard [10] and Tikhonov and Arsenin [26], and was made precise for statistical learning by De Vito et al. [5], who identify the learning problem with the inversion of a compact operator and derive regularization rates accordingly. The framing adopted here is the same in spirit and different in target. Those works invert towards a function; this one inverts towards an equivalence class, and the ill-posedness removed is not the instability of an operator inverse but the non-injectivity introduced by nuisance factors. Passing to the quotient does not regularize the problem, it changes the object recovered, and Proposition 2 states the condition under which this loses nothing diagnostically.

Invariance and geometric representation learning. Building known symmetries into a representation rather than learning them from data is the organizing idea of geometric deep learning [3], which classifies architectures by the group they respect. The nuisance factors of Section 3 form such a group, and the choice of a convolutional operator in Section 7 is made on that basis: translation equivariance matches the acquisition-phase nuisance. Where this work departs is that invariance is not asserted architecturally and then assumed. Section 9 reports a probe that tests whether a calibrated operator’s accepted region actually respects the geometry, and finds a linear operator that attains the target coverage while accepting points at hundreds of times the nominal spacing. Equivariance by construction constrains the representation; it does not by itself certify the decision region built on top of it.

Domain adaptation for fault diagnosis. A large recent literature transfers fault classifiers across operating conditions or machines by aligning feature distributions between a labelled source domain and an unlabelled target. The distinction from the present work is worth making precisely, because both are per-asset procedures and they are not the same per-asset procedure. Domain adaptation adapts the *representation* to each asset, and therefore needs target-domain data at training time, a criterion for when alignment has succeeded, and in the supervised variants labelled faults in the source domain. What is adapted here is not the representation but the *radius*: one operator is shared across assets, and each asset estimates a scalar from its own nominal data. The consequences are concrete. Adding an asset requires no retraining and no fault labels, the per-asset quantity has a coverage guarantee that a learned alignment does not, and the commissioning cost is a number of observations that can be stated in advance rather than a data requirement that can only be checked after the fact.

Topological and predictive representation learning. Two families are close enough in motivation to require explicit positioning. Self-organizing maps [13] also learn a topology-preserving organization of observations without labels, and joint-embedding predictive architectures [1] also learn representations by predicting in a latent space rather than reconstructing in the input space. Four differences separate them from the construction here. The organization is geometric over the learned representation rather than imposed on a fixed lattice, so the nominal region has an extent that can be measured rather than a grid coordinate. The problem is stated as an explicit inverse recovery with the invariance group named, rather than left implicit in the training objective. Convergence has a stopping criterion — saturation of the estimated radius, Definition 3 — rather than a fixed training budget. And labelling is attached to regions rather than to samples, which is what allows the taxonomy of Section 6 to grow without recalibrating classes already established.

Distribution-free tolerance limits and conformal prediction. The statistical result used in Section 7 is classical. Wilks [29] showed that order statistics of a sample give tolerance limits whose coverage does not depend on the

underlying distribution, and the modern development of the same exchangeability argument is conformal prediction [14, 15, 24, 28], which converts a nonconformity score into a prediction set with finite-sample validity.

We claim no contribution to that theory. The contribution is the identification: the per-asset diagnostic radius is exactly a distribution-free tolerance limit on the nominal residual, and the decision rule of Section 7 is a conformal predictor whose nonconformity score is the reconstruction residual. Once the identification is made, two things follow that are not usually stated in diagnostic terms. The attained coverage is known exactly rather than bounded (Proposition 7), so a commissioning procedure can be specified rather than tuned. And the guarantee has a minimum sample size (Corollary 6) together with a range above it in which the estimator is degenerate (Corollary 7) — a caveat that matters in practice, where commissioning data is scarce and the temptation is to certify from as few observations as the guarantee formally permits.

3. Observation Model

Let $\mathcal{X} \subset \mathbb{R}^{n_w}$ denote the space of fixed-length observation windows, where n_w is the window length. In the experiments of Sections 8–9 these observations are vibration windows, but the formulation does not depend on this particular sensing modality.

Let $\theta \in \Theta$ denote the latent physical state of the system. In a rolling-element bearing application, examples of θ include nominal operation, outer-race degradation, inner-race degradation, ball damage, or transitional states not yet assigned a semantic label.

Let $\eta \in \mathcal{E}$ denote nuisance variables that modify the observed signal without changing the underlying physical state. Typical nuisance variables include:

- amplitude scaling due to gain or load variation;
- phase shift due to sampling alignment;
- speed perturbation;
- DC offset or baseline drift;
- sensor placement and mounting effects;
- additive measurement noise.

The observation process is written as

$$x = F(\theta, \eta), \quad x \in \mathcal{X}. \quad (1)$$

Remark 1 (The observation window is a delay-coordinate vector). *The choice of a fixed-length window as the observation object is not a convention imposed for convenience. Writing the sampled signal as $\{s(t_0 + j/f_s)\}_{j=0}^{n_w-1}$, the window*

$$x = (s(t_0), s(t_0 + \Delta), \dots, s(t_0 + (n_w - 1)\Delta)), \quad \Delta = 1/f_s, \quad (2)$$

is exactly a delay-coordinate vector with lag Δ and embedding dimension $m = n_w$. If the underlying mechanical process is deterministic with a compact invariant set of box-counting dimension d , then by the embedding theorems of Takens [25] and Sauer et al. [19] the map (2) is generically an embedding whenever $m > 2d$, so $\mathcal{X}$ contains a diffeomorphic copy of that set and its topological invariants are preserved. For the bearing instantiation of Section 8.3, $m = 1200$ exceeds $2d$ by a very wide margin, which is why the representation operator of Section 7 is best understood as reducing a heavily redundant embedding rather than as extracting features from an unstructured vector.

Two qualifications are needed and are carried throughout. The embedding theorems require determinism, whereas (1) includes measurement noise; reconstruction under noise is a weaker and distinct theory [4], and the practical consequence is that the observed nominal set is a thickened neighbourhood of the reconstructed invariant set rather than the set itself. Under Assumption 2 that neighbourhood is bounded and the thickened set remains compact, which is what Section 7 needs; what is lost is the smooth manifold structure a strict embedding would supply, and it is accordingly not claimed (Remark 2).

Second, no result in this paper depends on the embedding reading. The compactness argument of Section 4 proceeds by equicontinuity and does not invoke it. Its role is to explain why the nominal set should be expected to have intrinsic

dimension far below n_w , which is what makes the covering argument of Remark 16 usable rather than vacuous, and it yields one quantitative prediction, tested in Section 9.

Physically, nuisance variables are constrained to bounded operating ranges: amplitude scaling, phase and speed perturbation, and offset all vary within finite intervals dictated by the installation and the operating envelope. We therefore record the following standing assumption, which is used in Section 4 to establish boundedness of the nominal functional structure.

Assumption 1 (Bounded nuisance range). *The nuisance parameter space $\mathcal{E}$ is a compact subset of a finite-dimensional parameter space, and for the nominal state θ_N the map $\eta \mapsto F(\theta_N, \eta)$ is uniformly bounded and equicontinuous on $\mathcal{E}$.*

A second boundedness assumption concerns the measurement rather than the operating point. It is used only in Section 7, but it belongs here because it is a statement about the observation process.

Assumption 2 (Bounded measurement perturbation). *The perturbation ϵ separating an observation from the deterministic orbit satisfies $\|\epsilon\| \leq \delta$ for a finite δ fixed by the acquisition chain.*

Physical measurement is bounded: the transducer, the anti-alias filter and the converter all have finite range, so an unbounded noise law is an analytic convenience rather than a claim about the instrument. The consequence drawn in Section 7 is that the observed nominal set, and not merely the deterministic orbit, is compact; the interpretive consequence, that a perturbation outside the calibrated budget is itself a departure from nominal operation, is recorded in Remark 17.

The diagnostic objective is not to identify η , nor is it necessarily to reconstruct the full forward model F , which may be unknown, nonlinear, high-dimensional, or impractical to identify. The diagnostic objective is to recover information about θ from x while remaining invariant to nuisance degrees of freedom. This produces the inverse problem

$$x \mapsto \theta, \quad (3)$$

or, more precisely, recovery of the physical state modulo nuisance transformations of the observation process. The inverse problem (3) is ill-posed in the classical sense [10, 26]: distinct nuisance configurations produce distinct observations corresponding to the same latent state, so the forward map is not injective in the relevant sense and no stable pointwise inverse exists without regularization.

4. Quotient-Space Formulation

The key mathematical object is the equivalence relation induced by nuisance transformations. Let $\mathcal{T}$ denote the physically motivated family of nuisance transformations introduced in Section 3: amplitude scaling, phase shift, speed perturbation, offset, and additive noise.

For the formal quotient argument, we let G_{nuis} denote the transformation group generated by $\mathcal{T}$, together with the identity transformation and the inverses of its elements. This is a closure construction: even if the physically motivated transformations in $\mathcal{T}$ are realized only over bounded operational ranges, the group generated by transformations of the same type is mathematically well defined.

The formal quotient-space construction is therefore stated in terms of the group action of G_{nuis} on $\mathcal{X}$. The compactness argument below does *not* require the full group G_{nuis} to be compact; it uses only the compact nuisance parameter set $\mathcal{E}$ of Assumption 1. Thus the mathematical quotient is defined using a genuine group action, while any empirical instantiation samples a compact, physically admissible subset of nuisance configurations. The consequences of this gap are revisited in Section 11.

Definition 1 (Nuisance equivalence). *Two observations $x_1, x_2 \in \mathcal{X}$ are nuisance-equivalent, written $x_1 \sim x_2$, if there exists $g \in G_{\text{nuis}}$ such that*

$$x_2 = g \cdot x_1, \quad (4)$$

and both observations correspond to the same latent physical state θ .

The orbit of an observation x under nuisance transformations is

$$[x] = \{g \cdot x : g \in G_{\text{nuis}}\}, \quad (5)$$

and the quotient space is the set of such equivalence classes:

$$\mathcal{X}/G_{\text{nuis}} = \{[x] : x \in \mathcal{X}\}. \quad (6)$$

We write $\pi : \mathcal{X} \rightarrow \mathcal{X}/G_{\text{nuis}}$ for the canonical projection $\pi(x) = [x]$, and equip $\mathcal{X}/G_{\text{nuis}}$ with the quotient topology, so that π is continuous by construction.

Definition 2 (Nominal quotient manifold). *The nominal quotient manifold $\mathcal{M}_{\text{N}}$ is the subset of $\mathcal{X}/G_{\text{nuis}}$ generated by observations sharing the nominal latent physical state:*

$$\mathcal{M}_{\text{N}} = \{[x] \in \mathcal{X}/G_{\text{nuis}} : x = F(\theta_{\text{N}}, \eta), \eta \in \mathcal{E}\}. \quad (7)$$

Proposition 1 (Compactness of the nominal nuisance orbit). *Under Assumption 1, the nominal nuisance orbit*

$$\mathcal{O}_{\text{N}} = \{F(\theta_{\text{N}}, \eta) : \eta \in \mathcal{E}\} \subset \mathcal{X}$$

is relatively compact: its closure $\overline{\mathcal{O}_{\text{N}}}$ is a compact subset of $\mathcal{X}$, where $\mathcal{X}$ is regarded as a space of fixed-length observation windows equipped with the uniform (sup-norm) topology.

Proof. By Assumption 1, the family $\{F(\theta_{\text{N}}, \eta)\}_{\eta \in \mathcal{E}}$ is uniformly bounded and equicontinuous in the sup-norm topology. By the Arzelà–Ascoli theorem [17, Thm. 7.25], such a family is relatively compact in that topology. Therefore $\overline{\mathcal{O}_{\text{N}}}$ is compact. $\square$

Corollary 1 (Boundedness and connectedness of the nominal quotient structure). *Assume the hypotheses of Proposition 1. Since the quotient projection π is continuous, the image $\pi(\overline{\mathcal{O}_{\text{N}}})$ is compact in $\mathcal{X}/G_{\text{nuis}}$. If, in addition, $\mathcal{E}$ is connected and $\eta \mapsto F(\theta_{\text{N}}, \eta)$ is continuous, then $\mathcal{O}_{\text{N}}$ is connected and hence so is $\pi(\mathcal{O}_{\text{N}}) = \mathcal{M}_{\text{N}}$. Consequently $\mathcal{M}_{\text{N}}$ is contained in a compact, connected subset of the quotient space.*

Proof. Continuous images of compact sets are compact and continuous images of connected sets are connected; $\mathcal{E}$ is a product of bounded intervals in the instantiation of Section 8, hence connected, and $\mathcal{M}_{\text{N}} = \pi(\mathcal{O}_{\text{N}})$ by Definition 2. $\square$

Remark 2 (On the use of the term “manifold”). *Proposition 1 and Corollary 1 establish compactness and connectedness properties of the nominal structure under explicit assumptions, but they do not establish a smooth or topological manifold structure in the differential-geometric sense, such as local Euclidean charts, a well-defined dimension, or curvature. The term functional manifold is used here, consistently with the companion report on manifold saturation [9] and with common usage in the representation-learning literature under the manifold hypothesis [3], to denote a bounded, connected region of the quotient space generated by a fixed latent regime under nuisance variability. A fully differential-geometric treatment is left to future work.*

Proposition 2 (Quotient factorization of diagnostic maps). *Assume that G_{nuis} acts on $\mathcal{X}$, that the action preserves the latent physical state, and that a diagnostic map $h : \mathcal{X} \rightarrow \mathcal{Y}$ depends only on that latent physical state. Then h is constant on G_{nuis} -orbits and therefore factors through the quotient projection π ; that is, there exists $\bar{h} : \mathcal{X}/G_{\text{nuis}} \rightarrow \mathcal{Y}$ such that*

$$h = \bar{h} \circ \pi. \quad (8)$$

Proof. Let $x_1, x_2 \in \mathcal{X}$ lie in the same G_{nuis} -orbit, so that $x_2 = g \cdot x_1$ for some $g \in G_{\text{nuis}}$. By assumption the action of g preserves the latent physical state, and h depends only on that state; hence $h(x_1) = h(x_2)$. Therefore h is constant on equivalence classes, and by the universal property of quotient maps it factors through π , yielding $\bar{h}$ with $h = \bar{h} \circ \pi$. $\square$

Corollary 2 (Equivalence-class diagnosis). *If two observations lie in the same nuisance orbit, no physically meaningful diagnostic decision distinguishes them on the basis of nuisance variation alone. Consequently, nominal observations should be identified as elements of a nominal equivalence class, or of a connected region of such classes, in the quotient space.*

Under this formulation, anomaly detection becomes a membership problem:

$$[x] \in \mathcal{M}_N \quad \text{or} \quad [x] \notin \mathcal{M}_N. \quad (9)$$

An anomaly is therefore not primarily a label. It is a failure of membership in the nominal quotient manifold. Classification, when required, is a subsequent operation that assigns non-nominal observations to other quotient-space regions or fault manifolds; this is developed in Section 6.

The compactness results describe the nominal set. The object a diagnostic actually has to locate, however, is its *boundary*, and it is worth recording that the boundary is as well behaved under the group action as the set itself.

Proposition 3 (The nominal boundary is nuisance-invariant). *Let G_{nuis} act on $\mathcal{X}$ by homeomorphisms and let $S \subseteq \mathcal{X}$ be G_{nuis} -invariant, that is $g \cdot S = S$ for every $g \in G_{\text{nuis}}$. Then the interior, the closure and the topological boundary $\partial S = \overline{S} \setminus \text{int } S$ are each G_{nuis} -invariant. In particular the nominal orbit $\mathcal{O}_N$ is G_{nuis} -invariant by construction, so $\partial \mathcal{O}_N$ is G_{nuis} -invariant and descends to a well-defined subset of $\mathcal{X}/G_{\text{nuis}}$.*

Proof. Each $g \in G_{\text{nuis}}$ acts as a homeomorphism, and homeomorphisms commute with closure and interior [17, Ch. 2]: $g \cdot \overline{S} = \overline{g \cdot S} = \overline{S}$ and $g \cdot \text{int } S = \text{int}(g \cdot S) = \text{int } S$. Hence $g \cdot \partial S = g \cdot \overline{S} \setminus g \cdot \text{int } S = \partial S$. Invariance under every g is exactly the condition for a subset to be a union of orbits, so $\partial \mathcal{O}_N = \pi^{-1}(\pi(\partial \mathcal{O}_N))$ and the boundary is the preimage of a subset of the quotient. $\square$

Proposition 3 is what makes “the boundary of normal operation” a meaningful target rather than a moving one. A change in an operating variable that belongs to the nuisance family, a different load, speed or temperature within the calibrated range, transports an observation *along* the boundary and leaves the boundary itself fixed; the invariants of the observation change while the set does not. A change that does not belong to that family does move the boundary, and degradation is precisely such a change. The two situations are therefore distinguishable in principle, and Section 10.5 returns to the distinction in dynamical terms.

Remark 3 (Scope of the invariance). *The proposition is a statement about a group action, so it is only as good as the identification of $\mathcal{T}$ with the transformations the physical process actually realizes. If an operating variable varies in a way not generated by $\mathcal{T}$, it is not quotiented, the nominal set is not invariant under it, and the boundary moves; the guarantee of Section 7 then degrades in the manner recorded in Section 11. Invariance is thus a modelling commitment about which variations are nuisance, not a property that can be read off the data.*

Remark 4 (The nuisance group acts on the reconstructed orbit). *Read through the embedding of Remark 1, the elements of $\mathcal{T}$ are not arbitrary signal manipulations but act recognizably on the reconstructed invariant set. A phase shift translates the delay vector along the trajectory and is therefore the time- δ map of the flow itself, so quotienting by phase is exactly the passage from a trajectory to the orbit regarded as a set. A speed perturbation is a reparametrization of that orbit; an amplitude change is a radial scaling of the embedded set; a constant offset is a translation along the diagonal of $\mathbb{R}^{n_w}$.*

This gives the quotient construction a dynamical reading: $\mathcal{M}_N$ is the reconstructed nominal orbit with the flow direction and the bounded reparametrization, scaling and offset directions collapsed, and what remains is the geometry that identifies the operating regime rather than the point along it. It also makes the intrinsic dimension of the nominal observation set predictable, since each nuisance parameter that acts non-trivially contributes one direction while noise contributes thickness rather than dimension. That prediction is quantitative and is tested in Section 9.

Remark 5 (Relation to the functional-topological foundation). *The compact, connected structure $\mathcal{M}_N$ obtained in Corollary 1 plays the role of the compact perceptual manifold of finite Hausdorff radius postulated in the functional-topological foundation of Di Santi [8], and the present section can be read as supplying one mechanism by which such a structure arises for a specific physical process. The two formal settings are, however, not identical and should not be conflated. In Di Santi [8], compactness is asserted directly for the set of valid realizations of a physical process, viewed as a subset of a Banach space of signals. Here, compactness is established for the nominal orbit in observation space (Proposition 1) and then transported to the quotient $\mathcal{X}/G_{\text{nuis}}$ by continuity of π (Corollary 1). The quotient construction is what makes the resulting structure low-complexity: nuisance directions are collapsed rather than merely bounded. Accordingly, the present work does not re-derive the general functional-topological claim; it specializes it to a nuisance group acting on a rotating-machinery observation process and supplies the additional geometric object, the quotient, that the diagnostic decision actually requires.*

5. Functional Manifolds and Blueprint Saturation

Section 4 established that the nominal structure is compact and connected. This section explains why that structure is the object a representation learner can plausibly recover, and makes the saturation criterion of Di Santi [9] precise enough to be certified rather than merely observed.

5.1. Functional manifolds as quotient geometry

The quotient-space formulation supplies a mechanism through which functional manifolds arise. A functional manifold is not postulated as an arbitrary structure that an algorithm happens to discover. It arises because nuisance transformations generate orbits in observation space, and the quotient projection collapses each orbit to a single equivalence class.

The observed structure therefore has two distinct sources:

- (i) physical continuity of the underlying system, which makes the orbit of a fixed latent regime a connected set; and
- (ii) identification of observations that differ only by nuisance transformations, which removes the directions along which that orbit is extended.

In raw signal space, nominal operation may appear as a broad and noisy cloud, because the nuisance directions inflate it in every coordinate that the sensor happens to be sensitive to. In quotient space those directions are collapsed, and what remains is closer to the structure of the latent physical state. This is the reason a representation that is insensitive to nuisance variability can expose compact functional regions where raw observations show none. The projection does not create physical structure; it removes variability that is irrelevant to the latent state.

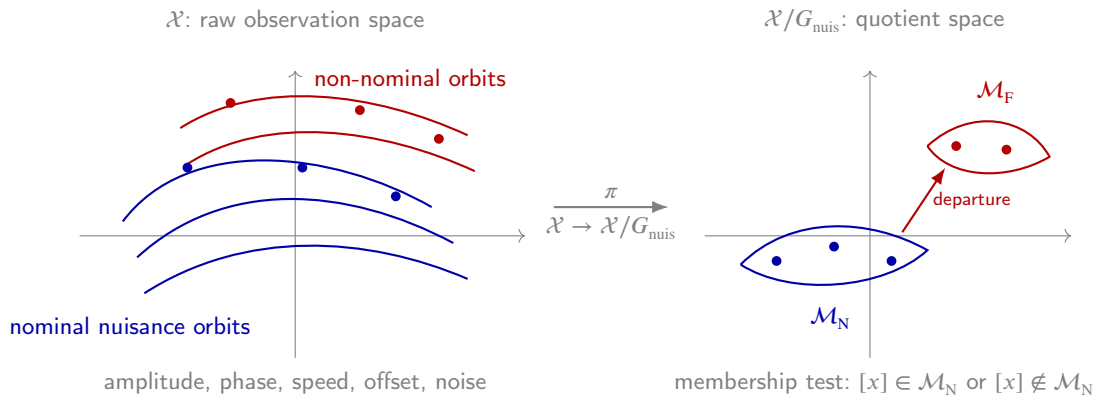


Figure 1: Functional manifolds as quotient geometry. In raw observation space, nuisance transformations generate extended orbits of observations that share the same latent physical state. The quotient projection π collapses those orbits into equivalence classes, exposing compact nominal and non-nominal functional regions. Diagnosis is then a membership test in $\mathcal{M}_N$ rather than a partition of labelled observations.

This also clarifies the relation between anomaly detection and classification. A supervised classifier learns a partition of labelled observations. A quotient-space diagnostic operator learns a representation in which physically equivalent observations are identified. Operational labels may subsequently be attached to the recovered regions, but the regions precede the labels; that is the subject of Section 6.

5.2. The saturation radius

The companion report [9] proposes that a regime becomes learnable when its geometric descriptors stabilize under additional observations, written informally as

$$H(n) \rightarrow H^*, \quad r_{\max}(n) \rightarrow r^*, \quad V(n) \rightarrow V^*, \quad (10)$$

for an entropy, a radius and a volume descriptor. As stated, (10) is a description of observed behaviour rather than a claim with a definite content: neither the estimators nor the mode of convergence is specified. We therefore fix one

descriptor precisely, the radius, and show that doing so brings saturation within reach of the same machinery used in Section 7.

Let $\phi : \mathcal{X} \rightarrow \mathcal{Z}$ be a representation map and let ν denote the law induced on $\mathcal{Z}$ by drawing η from the nominal nuisance law and applying $\phi \circ F(\theta_N, \cdot)$. Fix a reference point $c \in \mathcal{Z}$, for instance the mean of ν , and define the scalar

$$D = d(\phi(x), c), \quad x = F(\theta_N, \eta), \quad (11)$$

with distribution function F_D .

Definition 3 (Population and empirical saturation radius). *For $p \in (0, 1)$ the population p -radius of the nominal regime is the quantile*

$$R_p = F_D^{-1}(p) = \inf \{r \geq 0 : F_D(r) \geq p\}, \quad (12)$$

and given n nominal observations with radial values $D_1, \dots, D_n$ the empirical p -radius $\hat{R}_p(n)$ is the corresponding empirical quantile. The descriptor $r_{\max}(n)$ of (10) is the case $p \rightarrow 1$, and the p_{95} and p_{98} radii used in the companion work are the cases $p = 0.95$ and $p = 0.98$.

The first thing this definition buys is that the object being estimated exists and is finite, which does not follow from (10) on its own but does follow from Section 4.

Corollary 3 (Finiteness of the saturation radius). *Under Assumption 1, if ϕ is continuous then D is a bounded random variable and $R_p < \infty$ for every $p \in (0, 1)$; moreover $\sup_{\eta \in \mathcal{E}} D < \infty$, so the regime has a finite radius in the Hausdorff sense.*

Proof. By Proposition 1 the closure $\overline{\mathcal{O}_N}$ is compact, and ϕ is continuous, so $\phi(\overline{\mathcal{O}_N})$ is compact in $\mathcal{Z}$. The map $z \mapsto d(z, c)$ is continuous, and a continuous real function on a compact set is bounded and attains its supremum. Hence D is bounded and every quantile of F_D is finite. $\square$

Corollary 3 is the precise sense in which the compactness result of Section 4 underwrites the finite-Hausdorff-radius premise of the functional-topological foundation discussed in Remark 5. Saturation can then be stated as a convergence claim about an estimator of a finite quantity.

Proposition 4 (Consistency of the empirical saturation radius). *Let $D_1, D_2, \dots$ be independent draws of (11) and suppose F_D is strictly increasing in a neighbourhood of R_p , so that the quantile is unique. Then*

$$\hat{R}_p(n) \longrightarrow R_p \quad \text{almost surely as } n \rightarrow \infty. \quad (13)$$

Proof. By the Glivenko–Cantelli theorem the empirical distribution function converges to F_D uniformly and almost surely. Uniqueness of the quantile at level p transfers uniform convergence of the distribution function to pointwise convergence of the quantile function at p ; see, for example, Serfling [23, Sec. 2.3] or van der Vaart [27, Lemma 21.2]. $\square$

Proposition 4 makes (10) a theorem rather than an observation, and it identifies the limit as a specific population quantile. It is, however, purely asymptotic, and therefore silent on the question the operational criterion actually poses, namely how many observations suffice.

5.3. Finite-sample certification of saturation

That question is answered by the argument of Section 7.4. makes the saturation radius a quantile of a nominal distribution estimated from finitely many samples, which is exactly what the local operational radius is; only the space differs, distances in $\mathcal{Z}$ rather than residuals in $\mathbb{R}$. Proposition 7 is stated for an arbitrary exchangeable scalar sequence and therefore applies unchanged.

We note the forward dependency explicitly. Proposition 7 is proved in Section 7.4 rather than here because its operational role is the calibration of the decision boundary, which is where a reader will expect to find it; but it is a statement about exchangeable real-valued sequences and is logically independent of everything specific to residuals, as recorded in Remark 23. The corollary below therefore borrows a later result without circularity.

Corollary 4 (Distribution-free certification of the saturation radius). *Let $D_1, \dots, D_n, D_{n+1}$ be exchangeable with an atomless marginal, let $\alpha = 1 - p$, and set*

$$\hat{R}_p^\star = D_{(k_\alpha)}, \quad k_\alpha = \lceil p(n+1) \rceil. \quad (14)$$

Then, provided $n \geq 1/\alpha - 1$,

$$p \leq \mathbb{P}(D_{n+1} \leq \hat{R}_p^\star) = \frac{k_\alpha}{n+1} \leq p + \frac{1}{n+1}, \quad (15)$$

and $\hat{R}_p^\star$ equals the largest observed radial value $D_{(n)}$ precisely when $1/\alpha - 1 \leq n < 2/\alpha - 1$.

Proof. Apply Proposition 7, Corollary 6 and Corollary 7 with r_i replaced by D_i . Neither the coverage argument nor the two corollaries uses any property of the scalar sequence beyond exchangeability and atomlessness. $\square$

Corollary 4 unifies the two halves of the framework, and its practical content is the same in both. Certifying that a regime's empirical extent covers a fraction p of its nominal orbit requires at least $1/\alpha - 1$ observations, so 49 at $p = 0.98$; and because the certified radius is the observed maximum throughout $1/\alpha - 1 \leq n < 2/\alpha - 1$, a saturation curve that appears to have levelled off anywhere in that band has levelled off at the most variable statistic available, which is a property of the estimator rather than evidence about the regime. For $p = 0.98$ a non-degenerate saturation radius requires $n \geq 99$.

Remark 6 (Plug-in reference points break exchangeability). *The radial variable (11) is defined against a fixed reference c . In practice c is estimated from data, and the companion work additionally estimates the projection in which distances are measured from the same sample. Both substitutions make $D_1, \dots, D_n$ depend on one another and destroy the exchangeability of the augmented sequence, so Corollary 4 then holds only approximately, with an error that vanishes as the reference is estimated more precisely. The remedy is the same split-sample construction available in Remark 21: estimate the reference point, and the projection if one is used, on a block of nominal observations disjoint from the block used to compute the radius. This restores exchangeability by construction and costs only observations, not assumptions.*

Remark 7 (What a saturation curve does and does not establish). *Observed flattening of $\hat{R}_p(n)$ is a stopping heuristic, not a proof that R_p has been recovered, exactly as convergence of the commissioning criterion is not a proof that the population residual quantile has been recovered (Remark 24). An upper quantile estimated from few upper-tail observations can remain flat over an interval and then move again when a further extreme nominal observation arrives, and Corollary 4 says nothing to the contrary: it bounds coverage at a given n , not the increments of the curve. Claims of the form “the geometry has stabilized by $n \approx 500$ ” should therefore be read as descriptions of the observed trajectory, with the coverage statement supplying the part that is guaranteed.*

Remark 8 (The saturation curve and the guarantee pull in opposite directions). *The measurements reported in Section 9 sharpen Remark 7 into a concrete warning about the estimator, and the effect is not small. Inside the degenerate range the certified radius is the sample maximum, so it drifts systematically upward with n : on simulated nominal data at $p = 0.98$ its mean rises from about 2% above R_p at $n = 49$ to about 6% above at $n = 98$, and then falls discontinuously at $n = 99$, where an interior order statistic first becomes available. A saturation curve computed with the certified estimator therefore cannot flatten inside the band; apparent settling there is the maximum of a growing sample.*

The interpolated estimator behaves in exactly the complementary way. It produces the familiar smooth curve approaching R_p from below, and it undercovers at every sample size examined. The reassuring visual and the coverage guarantee are thus in tension: the estimator that makes a saturation plot look convergent is precisely the one that fails to certify, and the estimator that certifies does not produce a flat curve until the degenerate range has been left behind. Saturation should accordingly be reported with the attained coverage alongside the curve, and with $n \geq 2/\alpha - 1$.

Remark 9 (The saturation radius is a diagnostic, not a detector). *One further empirical observation limits what a saturation analysis can be asked to show. Measuring the certified radius of each simulated regime under a shared two-dimensional nominal projection separates them barely at all: the ratios of fault to nominal radius are between 0.98 and 1.04, that is, no separation. The reason is mechanical. A low-dimensional principal projection of a raw vibration window is dominated by the shaft harmonics, whereas the fault signature is a low-amplitude impulse train contributing almost nothing to the leading variance directions. The same simulator, the same nominal-only fitting and the same membership rule yield 99–100% detection when membership is tested through a reconstruction residual instead (Section 9).*

Two consequences follow. First, a saturation plot is evidence about the sampling of the nominal geometry, not about regime separability, and should not be cited for the latter. Second, this is the empirical reason the framework poses diagnosis through a residual, as in Section 7, rather than through a radius in a projection: Assumption 3 asks the score to track distance to the nominal manifold, and a leading-variance radius demonstrably does not.

6. Emergent Classification via Expert Feedback

Sections 4–5 concern a single regime: the nominal one. This section states what the same construction gives when several regimes have been recovered, which is the point at which the framework makes contact with classification. The answer is deliberately asymmetric: the geometry supplies a great deal about when a class should *not* be rejected, and very little about when a competing class should be, and that asymmetry is the substance of the section.

6.1. Two levels of classification

Classification in this framework occurs at two levels, and conflating them is the source of most of the confusion the inverse-problem view is meant to dispel.

The first is *structural* classification. A new observation is assigned to a recovered functional region, to an emerging region that has not yet saturated, to a transition between regions, or to none of them. This assignment requires no semantic labels at all: it is induced by the geometry of the quotient space and by the membership test (9).

The second is *operational* classification. Once a region has stabilized in the sense of Section 5, an expert or a downstream policy may attach meaning to it: normal operation, harmless variation, friction, heat, degradation, or a known fault regime. Classification is therefore not absent before labels exist; it first appears as geometric membership and only later becomes semantic.

6.2. Manifold-level labelling

Let $\{\mathcal{M}_j\}_{j=1}^k$ denote a family of recovered compact regions in $\mathcal{X}/G_{\text{nuis}}$, of which $\mathcal{M}_N$ is one. Labels are attached at the level of regions rather than of samples, through a map

$$L : \{\mathcal{M}_j\}_{j=1}^k \longrightarrow \mathcal{Y}, \quad L(\mathcal{M}_j) = y_j. \quad (16)$$

The contrast with supervised practice is the object being labelled. A supervised classifier consumes $x_i \mapsto y_i$ over many individual samples; here the expert supplies $\mathcal{M}_j \mapsto y_j$, once per recovered region. The labelling effort is proportional to the number of regimes, not to the number of observations.

Given a residual r_j for each region, in the sense of Assumption 3, and a radius τ_j calibrated for that region by Proposition 7, the admissible set for an observation is

$$\Gamma(x) = \{j \in \{1, \dots, k\} : r_j(x) \leq \tau_j\}, \quad (17)$$

and the reported diagnosis is

$$\hat{y}(x) = \begin{cases} L(\mathcal{M}_j), & \Gamma(x) = \{j\}, \\ \{L(\mathcal{M}_j)\}_{j \in \Gamma(x)}, & |\Gamma(x)| > 1 \text{ (ambiguous)}, \\ \text{unknown}, & \Gamma(x) = \emptyset. \end{cases} \quad (18)$$

Writing the rule as a set rather than an arg min is not a stylistic choice. It is what the calibration of Section 7 actually licenses, and it makes the two failure modes explicit rather than hiding them behind a nearest-region assignment: an observation may satisfy no region, which is the abstention (9) already anticipates, or it may satisfy several. The construction is the class-conditional, or Mondrian, form of a conformal predictor [24, 28], arrived at here from the quotient geometry rather than imported.

Corollary 5 (Per-class non-rejection). *Fix a region $\mathcal{M}_j$ and suppose the residuals r_j evaluated on observations from regime j satisfy Assumption 4, with τ_j calibrated from $n_j \geq 1/\alpha_j - 1$ such observations by (33). Then for a further observation x drawn from regime j ,*

$$\mathbb{P}(j \in \Gamma(x)) \geq 1 - \alpha_j. \quad (19)$$

Consequently the true regime is retained in the admissible set with probability at least $1 - \alpha_j$, simultaneously for every regime, without any assumption on the residual distributions and without any calibration data from competing regimes.

Proof. Apply Proposition 7 to the exchangeable sequence generated by r_j on regime j ; the event $\{j \in \Gamma(x)\}$ is precisely $\{r_j(x) \leq \tau_j\}$. The statement is per-regime and the regimes are calibrated independently, so no joint assumption is required. $\square$

Corollary 5 formalizes what the framework buys. Each region is calibrated only against its own nominal behaviour, so a new regime can be added after deployment without recalibrating the others, and a regime for which no data yet exists simply contributes nothing to $\Gamma(x)$. This is the sense in which the taxonomy may grow: adding $\mathcal{M}_{k+1}$ changes (17) by one term and leaves (19) untouched for $j \leq k$.

Several structural consequences follow, and they are the ones the companion report anticipates informally [9]: two regions may carry the same label when they are operational variants of one regime; a single nominal label may resolve into several regions when the underlying behaviour is heterogeneous; an unfamiliar observation need not be forced into the nearest known class; and expert effort attaches to regions rather than to samples.

6.3. What coverage does not give: separation

Corollary 5 bounds the probability of wrongly *excluding* the true regime. It says nothing whatever about wrongly *including* a competing one, and the distinction is not a technicality.

Remark 10 (Coverage is not discrimination). *Nothing in (19) prevents $|\Gamma(x)| > 1$. The acceptance regions $\{r_j \leq \tau_j\}$ are calibrated independently and may overlap arbitrarily; in the extreme, if two regions are geometrically indistinguishable under the available residuals, every observation from either is admitted by both, and (19) still holds exactly. A guarantee about non-rejection is therefore compatible with a diagnosis that is always ambiguous. Reducing $|\Gamma(x)|$ requires separation between regions, which is a property of the representation and must be established separately; it does not follow from the quotient construction of Section 4, from compactness, or from any calibration argument.*

Remark 11 (Two obstructions visible in the present construction). *The computational realization of Section 7 does not by itself supply that separation, for two distinct reasons, and the experiments of Section 9 exhibit both.*

First, a scalar residual is a many-to-one map. The rule (25) tests departure from the nominal region and reports a single number, from which the direction of departure cannot be recovered. Detection of a fault and identification of its type are therefore not the same problem in this construction, and the second requires k residuals, one per region, each with its own representation and its own calibration, not one residual with k thresholds.

Second, the low-dimensional radial diagnostic of Section 5 does not separate the regimes it describes. Measured on simulated data under a shared two-dimensional nominal projection, the ratio of fault radius to nominal radius lies between 0.98 and 1.04 (Section 9), that is, no separation at all, even though the same simulator and the same nominal-only fitting give 99–100% detection through the reconstruction residual. Saturation of a region and separability of that region from its neighbours are independent properties, and evidence for the former is not evidence for the latter.

This is the precise sense in which fault classification remains open in the present work, and it locates the gap left by the companion technical note [7] rather than closing it. What has been established is that diagnosis factors through the quotient (Proposition 2), so a correct classifier exists as a function of $[x]$ alone; that regions, once recovered, can be labelled and calibrated individually with distribution-free per-class validity (Corollary 5); and that abstention and ambiguity are reported rather than concealed (18). What has not been established is that any particular representation renders the non-nominal regions separable. Section 12 records the experiment that would test it.

7. Computational Realization and Local Calibration

Sections 3–6 characterize the diagnostic object: a compact, connected nominal region $\mathcal{M}_N$ in a quotient space, with anomaly defined as failure of membership in that region. That characterization is not yet an algorithm. Two quantities must be supplied before (9) can be evaluated on data: a computable surrogate for distance to $\mathcal{M}_N$, and a decision boundary on that surrogate.

This section supplies both. Section 7.1 introduces the representation operator and the residual used as a distance surrogate. Section 7.2 explains, as a consequence of Proposition 2, why the representation is transferable across assets while the numerical scale of the residual is not, which is the structural reason a single global decision boundary cannot be used. Sections 7.3 and 7.4 then give the calibration procedure and its guarantee: the central new result, Proposition 7, shows that a radius chosen as an appropriate order statistic of nominal commissioning residuals attains a prescribed

coverage level without any assumption on the residual distribution, and Corollary 6 converts this into a minimum commissioning length.

Throughout this section we index physical assets, or more generally independently commissioned monitoring environments, by k .

7.1. Representation operator and diagnostic residual

Let $E : \mathcal{X} \rightarrow \mathcal{Z}$ be an encoder and $D : \mathcal{Z} \rightarrow \mathcal{X}$ a decoder, with reconstruction

$$\hat{x} = D(E(x)), \quad (20)$$

and define the *diagnostic residual*

$$r(x) = \frac{1}{n_w} \|x - \hat{x}\|_2^2. \quad (21)$$

In the present realization E and D are the encoder and decoder of a one-dimensional convolutional autoencoder trained exclusively on nominal windows generated by Monte Carlo sampling of $\mathcal{E}$; autoencoders have long been used both as nonlinear representation learners [11] and as reconstruction-based anomaly detectors [18]. No non-nominal observation is used at any stage of representation learning.

Assumption 3 (Residual–distance proxy). *The diagnostic residual $r(x)$ is treated as a computable proxy that is monotonically related to distance to the nominal quotient manifold:*

$$r(x) \approx d([x], \mathcal{M}_N). \quad (22)$$

Assumption 3 is an operational hypothesis, not a consequence of the training objective. Minimizing mean-squared reconstruction error on nominal data does not by itself guarantee that r is isometric, or even monotonic, with respect to any particular metric on $\mathcal{X}/G_{\text{nuis}}$. The assumption is therefore evaluated empirically in Section 9, where the separation between nominal and non-nominal residual distributions is the relevant evidence, and it is listed among the limitations in Section 11.

The substantive claim is not that mean-squared error is the correct residual. It is that a residual can serve as an operational test for membership in the nominal quotient manifold. Any score with the qualitative property in (22) may be substituted: a latent likelihood, a Mahalanobis distance in $\mathcal{Z}$, an energy score, a diffusion or denoising score, a predictive error, or a physics-informed residual. The calibration theory of Section 7.4 is stated for an arbitrary residual and is therefore unaffected by that choice.

7.2. Why the representation transfers but its scale does not

Proposition 2 states that the ideal diagnostic map factors through the quotient. Write that ideal map as the membership predicate

$$h(x) = \mathbf{1}[x] \notin \mathcal{M}_N, \quad (23)$$

so that $h = \bar{h} \circ \pi$ by Proposition 2, and h is constant on nuisance orbits. The computable surrogate actually evaluated on data is

$$h_\tau(x) = \mathbf{1}[r(x) > \tau]. \quad (24)$$

Remark 12 (Invariance is inherited by the predicate, not by the score). *The surrogate (24) does not inherit the orbit-invariance of (23). The residual r is a function on $\mathcal{X}$, not on $\mathcal{X}/G_{\text{nuis}}$: it is not G_{nuis} -invariant, because the reconstruction error attained by a fixed pair (E, D) varies over an orbit even when the latent state does not. Consequently, for a fixed τ the level set $\{r = \tau\}$ is not the π -preimage of a subset of the quotient, and h_τ is not constant on orbits. What the quotient structure does guarantee, via Proposition 2, is that a correct diagnostic decision exists as a function of $[x]$ alone. Recovering it from r requires choosing τ so that the level set approximates $\partial\mathcal{M}_N$ under the nuisance law actually realized by the asset being monitored.*

Remark 12 has a direct architectural consequence, and it is the reason the calibration problem is posed per-asset rather than globally. The representation (E, D) encodes which directions of variation are nuisance and which are diagnostic; that information is a property of the asset family and is transferable. The numerical value of r additionally reflects the sensor coupling, mounting, operating point, preprocessing, and the particular optimization trajectory that produced (E, D) ; none of that is transferable. We therefore adopt the following separation:

- one representation operator, learned once from simulated nominal observations and thereafter held fixed;
- one scalar *local operational radius* τ_k^* per monitored asset, estimated from that asset's own nominal observations.

The decision rule for asset k at time t is then

$$\hat{y}(x_{k,t}) = \begin{cases} \text{anomalous,} & r(x_{k,t}) > \tau_k^*, \\ \text{nominal,} & r(x_{k,t}) \leq \tau_k^*. \end{cases} \quad (25)$$

This prediction is falsifiable and is tested directly in Section 9: transferring the residual *scale* across the simulation-to-field boundary should fail, while transferring the representation and re-estimating only τ_k^* should not.

7.3. Monte Carlo reference calibration

A reference boundary can be computed entirely in simulation. Let μ_{MC} and σ_{MC} denote the mean and standard deviation of r over held-out nominal Monte Carlo windows, and let $q \in (0, 1)$ denote a target nominal coverage level. A parametric reference is

$$\tau_{\text{MC}}(q) = \mu_{\text{MC}} + z_q \sigma_{\text{MC}}, \quad (26)$$

with z_q the corresponding standard-normal quantile; alternatively $\tau_{\text{MC}}(q)$ may be taken as the empirical q -quantile of the synthetic nominal residual distribution.

The level q is not part of the mathematical formulation. It is an operational risk parameter: lower values increase sensitivity at the cost of false alarms, higher values reduce false alarms at the cost of delayed detection of weak or incipient deviations.

Remark 13 (τ_{MC} is a reference, not a deployable radius). *By Remark 12, $\tau_{\text{MC}}(q)$ calibrates the residual scale of the simulator, not of any physical asset. It is reported here because it is the natural baseline and because its failure is informative, not because it is proposed as an operating point. The corresponding experiment is the direct-transfer condition of Section 9.*

7.4. Distribution-free calibration of the local operational radius

We now give the calibration procedure a guarantee. Before doing so it is worth being explicit about what is being estimated, because the scalar form of τ_k^* invites a misreading.

Remark 14 (A set is being estimated, not a threshold tuned). *The decision rule (25) compares a scalar to a scalar, but the object it defines is the region*

$$\hat{S}_k = \{x \in \mathcal{X} : r(x) \leq \tau_k^*\}, \quad (27)$$

a level set of the residual, and the guarantee below is a statement about the nominal probability mass that $\hat{S}_k$ contains. The task is therefore support estimation rather than threshold calibration: no trade-off between error types is being tuned, and no fault data is consulted to place the boundary. This is the problem posed by Devroye and Wise [6], who introduced support estimation precisely in order to decide whether a new measurement belongs to the support of a distribution observed so far, and it is the problem the one-class formulation of Schölkopf et al. [20] and the minimum-volume-set literature [22] address with different estimators.

The distinction matters for how the framework should be judged. A tuned threshold is answerable to the error rates it achieves on labelled faults; an estimated region with prescribed nominal coverage is answerable to whether it attains that coverage, which can be checked without any fault observation whatsoever. The residual supplies the family of candidate regions, indexed by τ ; Section 4 supplies the reason a bounded region exists to be estimated (Corollary 1) and the reason its boundary is a well-defined object in the quotient (Proposition 3); and the argument below selects a member of the family with a distribution-free coverage property.

Remark 15 (A level is estimated, not a shape). *The previous remark should not be read as a stronger claim than it is. The calibration selects one member of the one-parameter family $\{r \leq \tau\}_{\tau>0}$; the shape of that region is fixed entirely by r and is not estimated at all. Different residuals induce quite different geometries for the same nominal data: a reconstruction error against a linear subspace yields regions elongated along the retained directions, a Mahalanobis distance yields ellipsoids, a nearest-neighbour distance yields unions of balls in the manner of Devroye and Wise [6], and a kernel one-class construction [20] or a nonlinear autoencoder yields level sets with no closed-form description.*

Proposition 7 is indifferent between them: each attains the stated coverage, and coverage alone cannot distinguish a well-shaped region from a badly shaped one, exactly as Remark 10 observes for the multi-region case.

That shape is consequential rather than cosmetic is not a conjecture here. Section 9 reports that a radius in a two-dimensional projection, whose acceptance region is a ball, fails to separate the fault regimes at all, while the reconstruction residual computed from the same nominal data under the same calibration separates them at 79–100%. The nominal coverage is the same in both cases; only the geometry of the accepted region differs.

The honest position is therefore that this work certifies the level and inherits the shape. Estimating the shape is the province of minimum-volume-set estimation [22] and of the support-estimation constructions cited above, which choose the region rather than a level within a fixed family, and combining such an estimator with the distribution-free calibration used here is a natural continuation recorded in Section 12.

Before calibrating anything it is worth asking whether the object being estimated exists at all. It does, and for purely topological reasons.

Proposition 5 (Existence of a finite nominal radius). *Assume the hypotheses of Proposition 1, and let the representation ϕ and the residual r be continuous. Then*

- (i) $\phi(\overline{\mathcal{O}_N})$ is compact in $\mathcal{Z}$;
- (ii) the acceptance region $\hat{S} = \{r \leq \tau\}$ is closed in $\mathcal{X}$ for every τ ; and
- (iii) r attains a finite maximum $r^* = \max_{x \in \overline{\mathcal{O}_N}} r(x)$, so that $\{r \leq r^*\} \supseteq \overline{\mathcal{O}_N}$.

Proof. (i) A continuous image of a compact set is compact [17, Thm. 4.14]. (ii) $\hat{S} = r^{-1}((-\infty, \tau])$ is the preimage of a closed set under a continuous map, hence closed. (iii) A continuous real-valued function on a compact set attains its supremum [17, Thm. 4.16]; the inclusion is immediate from the definition of the maximum. $\square$

Part (iii) is the exact, noise-free counterpart of the statistical radius: a single finite level suffices to accept the entire nominal orbit, with no exceptions. It is the topological content of the premise that physically generated signals do not vary without bound, and it is stronger than Corollary 3, which asserts finiteness only for $p \in (0, 1)$.

Two objections to Proposition 5 suggest themselves, and both can be answered at the cost of an additional and physically mild assumption. A third cannot, and survives.

The first objection is that r^* is a supremum over a set never observed in full, so that any finite sample yields a maximum over a subset and underestimates it. This is true but quantitatively harmless, because the set in question is compact and of low dimension.

Proposition 6 (The sample maximum approximates the true maximum). *Let $K \subset \mathcal{X}$ be compact, let r be Lipschitz on K with constant L , and let $x_1, \dots, x_n \in K$ have fill distance*

$$h_n = \sup_{x \in K} \min_{1 \leq i \leq n} \|x - x_i\|. \quad (28)$$

Then

$$0 \leq \max_{x \in K} r(x) - \max_{1 \leq i \leq n} r(x_i) \leq L h_n. \quad (29)$$

Proof. The left inequality is immediate since the x_i lie in K . For the right, let x^* attain $\max_K r$, which exists by Proposition 5(iii), and choose x_i with $\|x^* - x_i\| \leq h_n$. Then $r(x^*) - r(x_i) \leq L\|x^* - x_i\| \leq L h_n$, and $\max_j r(x_j) \geq r(x_i)$. $\square$

Remark 16 (The rate is governed by the intrinsic dimension). *Proposition 6 is only useful if h_n shrinks at a usable rate, and this is where the embedding reading of Remark 1 does substantive work rather than interpretive work. For a compact set that is d -regular, sampled from a density bounded below on it, the fill distance satisfies $h_n = O((\log n/n)^{1/d})$ almost surely, by a standard covering argument. The exponent involves the intrinsic dimension d of the nominal set and not the ambient dimension n_w .*

The distinction is not academic here. The ambient dimension is 1200, which would make any covering-based rate useless, whereas the intrinsic dimension measured in Section 9 on the same simulator is close to four. The nominal

set is therefore coverable, and the gap in (29) closes at a rate governed by four rather than by twelve hundred. What the delay embedding contributes is precisely the reason to expect $d \ll n_w$, and Experiment 7 measures d rather than assuming it.

Two conditions are worth naming. The Lipschitz constant L is not known in practice, so (29) gives a rate rather than a usable numerical bound. And the commissioning stream must actually explore the nuisance range: samples concentrated in one corner of $\mathcal{E}$ have a large fill distance regardless of n . The second condition is related to, but not implied by, the exchangeability of Assumption 4.

The second objection is that Proposition 5 concerns the deterministic orbit, whereas observations carry measurement noise, so that the observed nominal law has unbounded support and no finite τ attains coverage one. This objection rests on a modelling convention rather than on physics.

Under Assumption 2, stated with the observation model in Section 3, the observed nominal set is contained in the δ -neighbourhood $\mathcal{O}_N \oplus B(\delta)$, which is compact because a compact set dilated by a closed ball is compact. Consequently Proposition 5 applies to the *observed* set and not only to the deterministic skeleton, and a finite level accepts every nominal observation rather than merely every nominal orbit point.

Remark 17 (What counts as noise and what counts as anomaly). *Assumption 2 has a consequence that is easy to mistake for a limitation. An observation that cannot be written as an orbit point plus a perturbation of size at most δ lies outside the nominal set by construction, and the framework reports it as such. It does not distinguish “unusually large noise” from “anomaly”, and on this formulation it should not: a perturbation outside the calibrated budget is a departure from nominal operation, whatever its physical origin.*

Two things follow. Calling such an observation anomalous is not the same as calling it a fault; that further attribution is the classification problem of Section 6, which the present work leaves open. And the burden the assumption imposes is honest and checkable: δ is a property of the acquisition chain, so a claim that the noise budget has been exceeded is a claim about the instrument, which can be audited independently of any diagnostic model.

The third objection stands, and no additional assumption removes it. Part (iii) guarantees that a finite level *contains* the nominal set; it says nothing about how much else that level set contains, so it constrains false alarms and not missed detections. Containment is not tightness. This is the gap identified in Remark 15 seen from the other side: compactness delivers a level, and the geometry of what that level encloses remains a property of r .

Remark 18 (Two routes, and why both are kept). *The paper now offers two arguments for a finite operational boundary, with different assumptions and different conclusions, and it is worth being explicit that neither subsumes the other.*

The topological route assumes bounded nuisance (Assumption 1), bounded measurement perturbation (Assumption 2) and a Lipschitz score. It concludes that an exact finite radius exists, accepts the entire nominal set, and is approximated by the sample maximum with an error decaying at a rate governed by the intrinsic dimension. The conclusion is strong; the assumptions are substantive and the Lipschitz constant is unknown.

The statistical route assumes only exchangeability (Assumption 4). It concludes that a prescribed coverage $1 - \alpha$ is attained exactly, for any distribution whatever, from an explicitly computable number of observations. The conclusion is weaker; the assumptions are far weaker and everything in it is computable.

They are complementary in a specific sense. The topological route explains why there is a boundary to estimate and why estimating it is not hopeless in dimension 1200; the statistical route delivers an estimate with a guarantee that does not depend on any of that structure being correctly specified. A reader who rejects the compactness modelling still has the coverage result; a reader who accepts it gains an account of why the procedure works as well as it does with so few observations.

Remark 19 (A radius that would live in the quotient). *Proposition 3 places the nominal boundary in the quotient, yet the acceptance region (27) does not live there: by Remark 12 the residual is not G_{nuis} -invariant, so $\{r \leq \tau\}$ is not a union of orbits and does not descend under π . The asymmetry is worth naming, because it can be removed.*

Define the orbit-wise reduction

$$\bar{r}([x]) = \inf_{g \in G_{\text{nuis}}} r(g \cdot x), \quad (30)$$

which is constant on orbits by construction and therefore a function on $\mathcal{X}/G_{\text{nuis}}$. If the acting group is compact and r is continuous, the infimum is attained and $\bar{r}$ is continuous on the quotient, so $\{\bar{r} \leq \bar{\tau}\}$ is a closed subset of $\mathcal{X}/G_{\text{nuis}}$ and the calibration of Section 7.4 applies to it unchanged, the exchangeability argument being indifferent to which scalar

is being quantiled. In that formulation the radius is genuinely an object of the quotient rather than of the observation space, which is what the geometric framing of Sections 4 and 5 would lead one to expect.

This is not what is implemented here. Evaluating (30) requires minimizing over the group for every observation, which the present pipeline does not do and which for the transformation family of Section 8.3 would mean optimizing over amplitude, phase, speed and offset per window. The per-asset radius used throughout is instead an empirical accommodation of the same non-invariance: rather than removing the residual's dependence on the orbit representative, it absorbs that dependence into a scalar estimated locally. The two are different constructions with different costs, and comparing them is recorded in Section 12.

The construction that follows is a distribution-free one-sided tolerance limit, in the sense introduced by Wilks [29] and developed in the modern conformal-prediction literature [14, 15, 28]. Its relevance here is that the local operational radius τ_k^* is exactly such a tolerance limit on the nominal residual distribution of asset k , so that $\hat{S}_k$ is a distribution-free tolerance region for the nominal regime; the correspondence supplies both a coverage statement and a minimum commissioning length.

During commissioning, asset k acquires a finite sequence of observations believed to be nominal, and the shared representation converts them into residuals

$$r_1, r_2, \dots, r_n, \quad r_i = r(x_{k,i}). \quad (31)$$

Let $r_{(1)} \leq r_{(2)} \leq \dots \leq r_{(n)}$ denote the corresponding order statistics, and let r_{n+1} denote the residual of the next nominal observation from the same asset, that is, the first observation to be evaluated under the frozen radius.

Assumption 4 (Exchangeability of nominal commissioning residuals). *The residuals $r_1, \dots, r_n, r_{n+1}$ arising from nominal operation of asset k are exchangeable: their joint distribution is invariant under permutation of the indices. In addition, their common marginal distribution is atomless, so that ties occur with probability zero.*

Exchangeability is strictly weaker than independence and does not require the nominal residual distribution to be known, parametric, or even identified. It does require that commissioning and the immediately subsequent monitoring period be drawn from the same nominal regime; it is therefore violated by drift, by a non-stationary run-in transient, or by an asset that is already degraded at installation. These are precisely the failure modes recorded in Section 11, and Remark 22 below identifies the experiment that probes the assumption empirically.

Proposition 7 (Distribution-free coverage of the local operational radius). *Let Assumption 4 hold, let $\alpha \in (0, 1)$ be a target miscoverage level, and suppose the commissioning length satisfies*

$$n \geq \frac{1}{\alpha} - 1. \quad (32)$$

Define the local operational radius as the order statistic

$$\tau_k^* = r_{(k_\alpha)}, \quad k_\alpha = \lceil (1 - \alpha)(n + 1) \rceil. \quad (33)$$

Then $k_\alpha \leq n$, so (33) is well defined, and

$$1 - \alpha \leq \mathbb{P}(r_{n+1} \leq \tau_k^*) = \frac{k_\alpha}{n + 1} \leq 1 - \alpha + \frac{1}{n + 1}. \quad (34)$$

In particular the false-alarm probability on the next nominal observation satisfies $\mathbb{P}(r_{n+1} > \tau_k^) \leq \alpha$.*

Proof. We first verify feasibility. Since n is an integer, $\lceil (1 - \alpha)(n + 1) \rceil \leq n$ holds if and only if $(1 - \alpha)(n + 1) \leq n$, which rearranges to $1 \leq \alpha(n + 1)$, that is, to (32). Hence $k_\alpha \in \{1, \dots, n\}$ and $r_{(k_\alpha)}$ is a well-defined order statistic of the commissioning sample.

Define the rank of the future residual within the augmented sample,

$$R = 1 + \#\{i \in \{1, \dots, n\} : r_i < r_{n+1}\}. \quad (35)$$

By Assumption 4 the marginal is atomless, so almost surely the $n + 1$ residuals are distinct and R is the position of r_{n+1} in the increasing ordering of $r_1, \dots, r_{n+1}$.

We claim that, almost surely,

$$\{r_{n+1} \leq r_{(k_\alpha)}\} = \{R \leq k_\alpha\}. \quad (36)$$

Indeed, $r_{(k_\alpha)} < r_{n+1}$ holds if and only if at least k_α of the first n residuals are strictly smaller than r_{n+1} , since $r_{(k_\alpha)}$ is the k_α -th smallest of those n values. By (35) this is equivalent to $R \geq k_\alpha + 1$. Taking complements gives (36).

It remains to compute the law of R . Exchangeability of $(r_1, \dots, r_{n+1})$ implies that the joint distribution is invariant under any permutation of the indices; combined with almost-sure distinctness, this implies that r_{n+1} is equally likely to occupy any of the $n + 1$ positions in the increasing ordering. Hence R is uniformly distributed on $\{1, \dots, n + 1\}$ and

$$\mathbb{P}(r_{n+1} \leq \tau_k^\star) = \mathbb{P}(R \leq k_\alpha) = \frac{k_\alpha}{n + 1}. \quad (37)$$

Finally, the definition of the ceiling gives $(1 - \alpha)(n + 1) \leq k_\alpha < (1 - \alpha)(n + 1) + 1$, and dividing by $n + 1$ yields the two outer inequalities in (34). $\square$

Three features of Proposition 7 matter for the application. First, no assumption is placed on the shape of the nominal residual distribution, which is essential here because that distribution is the output of a learned nonlinear operator and is neither Gaussian nor otherwise tractable. Second, the guarantee is per-asset and requires only that asset's own nominal data, which is what makes the shared-representation architecture of Section 7.2 admissible. Third, the coverage is two-sided: (34) bounds the attained level from above as well as below, so the procedure is not needlessly conservative, with the excess vanishing at rate $1/(n + 1)$.

Corollary 6 (Minimum commissioning length). *Under Assumption 4, attaining distribution-free coverage $1 - \alpha$ by (33) requires*

$$n \geq \left\lceil \frac{1}{\alpha} \right\rceil - 1. \quad (38)$$

For the target coverage $1 - \alpha = 0.98$ used throughout this work, the minimum commissioning length is $n \geq 49$ observations, and (33) selects $k_\alpha = 49$ at $n = 49$, that is, the sample maximum.

Proof. Immediate from (32): if $n < 1/\alpha - 1$ then $k_\alpha > n$ and no order statistic of the commissioning sample attains the required level. Since n is an integer the condition is $n \geq \lceil 1/\alpha \rceil - 1$. Substituting $\alpha = 0.02$ gives $n \geq 49$, and $k_\alpha = \lceil 0.98 \cdot 50 \rceil = 49$. $\square$

Corollary 6 is worth emphasizing because it is a hard information-theoretic floor rather than a tuning choice: below 49 nominal observations, a 98% nominal-coverage radius cannot be certified from that asset's data alone by any distribution-free procedure, irrespective of the estimator used.

The floor also has a consequence that is easy to miss and that turns out to matter more in practice than the floor itself.

Corollary 7 (Degenerate range: the certified radius is the sample maximum). *Under the hypotheses of Proposition 7, $k_\alpha = n$, and hence $\tau_k^\star = r_{(n)} = \max_i r_i$, if and only if*

$$\frac{1}{\alpha} - 1 \leq n < \frac{2}{\alpha} - 1. \quad (39)$$

At $\alpha = 0.02$ this is $49 \leq n \leq 98$; an interior order statistic is obtained only from $n \geq 99$.

Proof. $k_\alpha = \lceil (1 - \alpha)(n + 1) \rceil = n$ holds if and only if $n - 1 < (1 - \alpha)(n + 1) \leq n$. The right inequality rearranges to $1 \leq \alpha(n + 1)$, that is $n \geq 1/\alpha - 1$, which is (32). The left inequality rearranges to $\alpha(n + 1) < 2$, that is $n < 2/\alpha - 1$. $\square$

Corollary 7 says that for every commissioning length in a band roughly twice the width of the floor, the certified radius is the sample maximum, which is the order statistic of largest variance. The practical implication is stronger than the floor: satisfying (38) is necessary but not sufficient for a stable radius, and a commissioning length chosen just above the floor purchases coverage at the cost of maximal estimator variance. Monte Carlo verification across six residual distributions (Section 9) gives a standard deviation of $\tau_k^\star$ that is essentially flat over $n \in [49, 98]$, falls by a factor of about 1.6 at the first interior order statistic $n = 99$, and by a factor of about 2.5 by $n = 200$. For a target coverage of 98% the recommendation is therefore $n \geq 99$, not $n \geq 49$.

Remark 20 (Interpolated quantiles are anti-conservative). *Standard empirical-quantile routines return a value obtained by linear interpolation between adjacent order statistics, which for an upper quantile lies below the order statistic $r_{(k_\alpha)}$ prescribed by (33). Such an estimate therefore does not satisfy (34), and the resulting false-alarm rate is systematically above the nominal level α . This is a concrete, testable prediction: any implementation that calibrates with an interpolated p_{98} rather than with the order statistic $r_{(k_\alpha)}$ should exhibit a positive bias in false-alarm rate across all assets, in the same direction in every case. Monte Carlo evaluation over six residual distributions (Section 9) confirms this: the interpolated estimator loses 1.9 percentage points of coverage relative to $r_{(k_\alpha)}$, consistently in sign and magnitude and irrespective of the distribution, whereas the certified estimator attains the coverage $50/51 = 0.9804$ predicted at $n = 50$ to within Monte Carlo error. Of the two bias sources considered here, interpolation is by an order of magnitude the dominant one (cf. Remark 21). The correction implied by Proposition 7 is to replace the interpolated quantile by $r_{(k_\alpha)}$ and to enforce (38).*

Remark 21 (Data-dependent stopping breaks exactness, but is conservative here). *Proposition 7 treats n as fixed in advance. The stopping rule of Section 7.5 makes n a function of the observed residual sequence, and an adaptively chosen stopping time destroys the exchangeability of the augmented sample $(r_1, \dots, r_n, r_{n+1})$, so the coverage statement holds approximately rather than exactly. Two distinct effects must be separated, and they act in opposite directions.*

The first is a genuine selection effect: a rule that terminates when the upper-tail estimate has been locally stable tends to stop at instants of quiet tail behaviour, which biases $r_{(k_\alpha)}$ downward and is therefore anti-conservative. The second is a sample-size effect: the rule does not stop at the floor but typically well above it, and by Corollary 7 the certified radius is the sample maximum throughout the relevant range, so a larger n yields a larger radius and higher coverage.

Monte Carlo evaluation (Section 9) shows the second effect dominates. Comparing the adaptive rule against a fixed- n calibration at the same realized n , which isolates selection from sample size, gives a coverage difference of at most 0.2 percentage points with inconsistent sign across residual distributions; the selection bias is therefore negligible at this configuration. Comparing the adaptive rule against a fixed $n = 50$ instead gives a coverage gain of about 0.6 percentage points, attributable to the larger realized n . Under the configuration used here, adaptive stopping is thus mildly conservative rather than anti-conservative. This should not be read as a general guarantee, since it depends on the stopping parameters and on operating inside the degenerate range of Corollary 7. Two remedies restore exactness if required: fix n in advance subject to (38), or use the stopping rule only to declare readiness and compute τ_k^ from a disjoint block of nominal observations.*

Remark 22 (The robustness ablation is an exchangeability probe). *Assumption 4 is testable in a specific sense. If the nominal commissioning residuals are exchangeable, then permuting their arrival order is distributionally inconsequential, and the calibrated radius, the convergence point, and the post-calibration false-alarm rate should vary only by finite-sample noise. The repeated-commissioning experiment of Section 9, in which the shared representation and all calibration parameters are held fixed while the commissioning observations and their order are permuted, is therefore not merely a robustness check: it is the natural empirical probe of the one assumption on which Proposition 7 rests. Systematic dependence on arrival order would falsify exchangeability and indicate drift within the commissioning window.*

Remark 23 (Scope: the proposition is not about residuals). *Nothing in Proposition 7 or in Corollaries 6 and 7 uses any property of r beyond exchangeability and atomlessness of the scalar sequence it generates. The results are therefore statements about exchangeable real-valued sequences, and they apply equally to any other nominal quantity estimated by an upper quantile. This is what licenses Corollary 4 of Section 5.3, where the same argument is applied to radial distances in representation space rather than to residuals, yielding the same floor and the same degenerate range. The saturation radius of a functional regime and the local operational radius of an asset are one estimation problem posed in two spaces.*

7.5. Convergence-based stopping and freezing

Corollary 6 gives a lower bound on the commissioning length but not a stopping time. In deployment the relevant question is when the estimated radius has stabilized enough to be frozen. Let

$$\tau_k(t) = \hat{Q}_{k,t}(1 - \alpha) \quad (40)$$

denote the estimate after t commissioning residuals, where $\hat{Q}_{k,t}$ is the empirical quantile function of $r_1, \dots, r_t$; by Proposition 7 the certified choice is $\hat{Q}_{k,t}(1 - \alpha) = r_{(\lceil (1-\alpha)(t+1) \rceil)}$. Define the relative variation over a look-back interval L ,

$$C_k(t) = \frac{|\tau_k(t) - \tau_k(t - L)|}{\max\{|\tau_k(t - L)|, \epsilon\}}, \quad (41)$$

with $\epsilon > 0$ a numerical stabiliser. Commissioning terminates at the first $t \geq N_{\min}$ for which $C_k(t) < \epsilon_C$ holds for m consecutive updates, at which point $\tau_k^* = \tau_k(t)$ is frozen and monitoring under (25) begins. If no such t is reached before a maximum budget $N_{\max}$, the asset is assigned a calibration-failure status and is not released into active monitoring.

Two constraints link this rule to the theory of Section 7.4. First, $N_{\min}$ must satisfy (38), since otherwise a converged radius can be declared at a sample size at which the target coverage is not certifiable; by Corollary 6 this requires $N_{\min} \geq 49$ at $\alpha = 0.02$, and strictly greater than 49 if the degenerate sample-maximum case is to be avoided. Second, by Remark 21 the rule introduces a downward bias in τ_k^* , so the parameters $(N_{\min}, N_{\max}, L, \epsilon_C, m)$ define a trade-off between commissioning duration and the size of that bias rather than a consistency property.

Remark 24 (What the stopping rule does and does not claim). *Convergence of (41) is not evidence that the population $(1 - \alpha)$ -quantile has been recovered. It is an operational statement that the estimate has been stable over the configured look-back under the configured tolerance. Non-convergence within $N_{\max}$ is correspondingly diagnostic rather than merely inconvenient: it indicates insufficient nominal data, non-stationary nominal operation, or startup behaviour inconsistent with the assumed nominal state, that is, a failure of Assumption 4. Distinguishing among these causes lies outside the present evaluation.*

7.6. Summary of the procedure

The realization consists of an offline stage performed once per asset family and an online stage performed once per asset.

Offline (once per asset family).

1. Sample nuisance configurations $\eta_i \in \mathcal{E}$ and generate nominal observations $x_i = F(\theta_N, \eta_i)$, thereby approximating the nominal orbit $\mathcal{O}_N$ of Proposition 1.
2. Train (E, D) on these nominal observations only, and freeze it.
3. Optionally record the synthetic reference $\tau_{MC}(q)$ of (26) as a baseline, subject to Remark 13.

Online (once per asset, at commissioning).

1. Acquire nominal residuals $r_1, r_2, \dots$ using the frozen (E, D) .
2. Update $\tau_k(t)$ by (40) and monitor $C_k(t)$ of (41), with $N_{\min}$ satisfying (38).
3. On convergence, freeze τ_k^* and begin monitoring under (25); on exhaustion of $N_{\max}$, withhold the asset with calibration-failure status.

No fault labels, no field representation learning, and no manual threshold selection occur at any point. The only asset-specific quantity is the scalar τ_k^* , and by Proposition 7 it carries a distribution-free coverage guarantee.

8. Experimental Design

The experiments are organized by the kind of evidence they supply rather than by increasing realism, because the three kinds are not interchangeable and a reader is entitled to know which claim rests on which.

The first kind is *verification*. Propositions 7 and 4 and their corollaries are mathematical statements; the appropriate check is whether the stated identities hold, not whether a dataset behaves well. These experiments use no data at all, and one of them uses no sampling either.

The second kind is *controlled instantiation*. A physics-motivated simulator supplies a setting in which the latent regime, the nuisance family and the departures from nominal are all known by construction, so the end-to-end chain of Section 7 can be exercised against ground truth.

The third kind is *field transfer*. Two public bearing benchmarks test whether a representation learned entirely in simulation retains diagnostic meaning on real measurements: the Case Western Reserve University (CWRU) dataset under controlled operating-condition variability, and the NASA Intelligent Maintenance Systems (IMS) run-to-failure dataset under naturally evolving degradation.

8.1. Design invariants

Four constraints hold throughout and are stated once here rather than repeated per experiment.

1. **No fault observation is used for fitting or calibration.** Fault data enters only at evaluation. This applies to the representation operator, to the reference threshold τ_{MC} , and to every operational radius.
2. **No field observation is used for representation learning.** The shared representation is trained on simulated nominal windows and transferred unchanged; CWRU and NASA IMS observations are used only to estimate a scalar radius and to evaluate.
3. **Calibration observations are disjoint from evaluation observations.** Reported false-alarm rates are computed on nominal observations not used to compute the radius, so coverage is an out-of-sample quantity.
4. **Every reported quantity is seeded.** Sampling, permutation and model initialization are all deterministic given the seed recorded with the results.

8.2. Verification of the calibration theory

Four checks, in decreasing order of strength.

Exhaustive enumeration. For small n the coverage identity of Proposition 7 is evaluated by enumerating all $(n + 1)!$ orderings of $n + 1$ distinct exchangeable values and counting the fraction in which the last falls at or below $r_{(k_a)}$. This is a check of the proposition itself rather than of a Monte Carlo approximation to it, and it is run over a grid of α .

Heterogeneous residual families. Proposition 7 is distribution-free, so the appropriate empirical test is not the residual distribution of one trained model but a family chosen to be as dissimilar as possible: lognormal, gamma, exponential, Pareto, half-normal and a bimodal mixture, spanning light and heavy tails, bounded and unbounded support, and unimodal and bimodal shapes. Any systematic dependence of attained coverage on the member of the family would falsify the proposition.

The estimator–stopping design. The two conventions of Section 7.4 are crossed with two stopping rules, giving four arms:

	adaptive stopping ($N_{\min} = 40$)	fixed $n = 50$
interpolated p_{98}	A1	A3
certified $r_{(k_a)}$	A2	A4

A1 reproduces the published pipeline exactly and A4 is the configuration to which Proposition 7 applies without qualification. Comparing the four cells separates the contribution of the estimator from that of the stopping rule, which the published configuration confounds. Because the A2–A4 contrast still mixes the stopping *selection* effect with the larger sample size the adaptive rule collects, a further control compares the adaptive rule against a fixed- n calibration on an independent block of the *same* realized size.

The exchangeability test. Assumption 4 is probed by the permutation test described in Remark 22, whose statistic is the absolute Spearman correlation between residual value and arrival index. Its size is measured under exchangeable streams and its power under an injected monotone drift of increasing magnitude, before it is applied to any real commissioning stream: a test with inflated size would make the robustness ablation look like evidence of a violated assumption, and a test without power would make it uninformative.

8.3. Controlled instantiation: the bearing simulator

The simulator generates windows of $n_w = 1200$ samples at $f_s = 12$ kHz, corresponding to 100 ms, with a nominal shaft frequency of 30 Hz. The nominal signal is a five-term harmonic expansion with geometric decay 0.55, plus a damped structural resonance at 3 kHz and broadband noise. It is not offered as a high-fidelity twin of any particular assembly; its purpose is to provide a regime with known latent structure, controlled nuisance variability and controlled departures.

Nuisance variables are drawn independently per window: amplitude $a \sim U(0.85, 1.15)$, phase $\phi \sim U(-0.3, 0.3)$, speed $s \sim U(0.95, 1.05)$, offset $\sim U(-0.05, 0.05)$, additional noise $\sim U(0, 0.03)$ and a short local dropout. These instantiate the compact nuisance set $\mathcal{E}$ of Assumption 1. Non-nominal regimes add impulse trains at 108s, 162s and 72s Hz for outer-race, inner-race and ball faults respectively, the last with a low-frequency amplitude modulation.

Three measurements are taken in this setting. The end-to-end chain of Section 7 is exercised with a representation operator fitted on nominal windows only, verifying that nominal residuals concentrate, that fault residuals do not, that a single membership test detects every fault family, and that assets differing only by an installation signature identify materially different radii. The saturation radius of Section 5 is measured against a large-sample reference R_p , with its dispersion, its relative bias and its attained coverage reported as functions of n across and beyond the degenerate range. Finally the same radius is measured per regime under a shared nominal projection, to test whether saturation of a region implies separability from its neighbours.

8.4. Field transfer: CWRU

The shared representation is transferred unchanged to the CWRU Drive-End dataset, sampled at 12 kHz, so that the 1200-sample window again corresponds to 100 ms. Four operating conditions are treated as independent deployment environments: 1730, 1750, 1772 and 1797 RPM, comprising 1,414 healthy, 1,621 ball-fault, 1,621 inner-race-fault and 2,846 outer-race-fault windows.

Three calibration strategies are compared under the identical representation, so that only the information available for estimating the radius changes:

1. **Direct threshold transfer**, applying the simulator-derived τ_{MC} unchanged. This is the empirical test of Remark 12: the representation is predicted to transfer and its numerical scale is not.
2. **Field oracle**, computing the radius retrospectively from the complete healthy record of each condition. Operationally unavailable; included as an upper reference.
3. **Online calibration**, estimating the radius sequentially under the stopping rule of Section 7.5.

A separate reference is provided by independently trained asset-specific detectors, one representation and one preprocessing transformation fitted per condition on its complete healthy record. This reference is deliberately favourable and operationally unrealistic; its role is to bound what is lost by sharing one representation.

The initialization-robustness ablation permutes the commissioning observations and their arrival order while holding the representation and all calibration parameters fixed. Following Remark 22, this is reported not merely as a robustness check but as the empirical probe of Assumption 4.

8.5. Field transfer: NASA IMS

A second representation is constructed for the IMS bearing family from simulated nominal observations matched to that dataset's sampling and operating characteristics, and transferred unchanged to each of four independently monitored bearings from Experiment 2, each comprising 984 chronologically ordered recordings at a 10-minute interval.

Each bearing estimates its radius from an early-life segment and freezes it. Departure is declared by a persistence rule over a fixed window of recordings, and the reported quantities are the number of observations to convergence, the spread of the converged radii across bearings, the healthy-period false-alarm rate, the time of first persistent departure, and the lead time to the documented end of the run.

Unlike CWRU, this setting has no discrete fault classes and no balanced evaluation set; what it tests is whether an independently commissioned radius remains diagnostically meaningful across a complete degradation trajectory without further adaptation.

8.6. Estimator conventions, and a caveat on the field results

One point must be declared here rather than allowed to surface in Section 9. The field experiments as conducted use the *interpolated* empirical p_{98} , with $N_{\min} = 40$, $N_{\max} = 200$, look-back $L = 10$, tolerance $\varepsilon_C = 0.01$ and $m = 10$ consecutive stable updates. In the terminology of Section 8.2 they are the A1 arm.

Two consequences follow, and both are consequences of results proved above rather than of any defect discovered in the data. First, by Remark 20 an interpolated upper quantile lies at or below $r_{(k_a)}$ and therefore does not satisfy the coverage bound, so the field radii do not carry the guarantee of Proposition 7; the systematic direction of the resulting false-alarm excess is predicted, and is examined in Section 9. Second, $N_{\min} = 40$ lies below the commissioning floor $1/\alpha - 1 = 49$ of Corollary 6, and the commissioning lengths the rule actually returns lie in $[49, 98]$, which by Corollary 7 is precisely the range in which the certified radius degenerates to the sample maximum.

Claiming the guarantee in the field therefore requires re-running calibration with $r_{(k_a)}$ and $N_{\min} \geq 2/\alpha - 1 = 99$, reporting both configurations. That comparison is inexpensive, since it operates on the residual streams alone and requires no retraining, and it is the natural extension of the present work; it is recorded in Section 12.

Remark 25 (A numerical coincidence that must not be read as evidence). *The commissioning lengths returned in the field cluster at 49, and the coverage floor of Corollary 6 at $\alpha = 0.02$ is also 49. These arise from unrelated mechanisms. The stopping rule cannot emit a calibration size below $\max\{N_{\min} - 1, \text{warm-up} - 1 + L\} + m$, which equals 49 at the configuration used and is independent of α ; the coverage floor is $1/\alpha - 1$ and does not involve the stopping parameters. The observed clustering is therefore uninformative about Corollary 6. Disentangling the two requires moving one floor without the other, for instance by setting $m = 5$, which moves the stopping floor to 44 while leaving α untouched.*

8.7. Declared uncertainties and reproducibility

Two details of the NASA IMS protocol are recorded as unresolved rather than silently fixed, because both affect reported lead times. The persistence rule appears in the source code with a threshold of three exceedances in five recordings, while the accompanying text describes eight in ten; and the recording-level score is computed as the mean of the window residuals in the code, while a 95th-percentile aggregation is also defined. The results in Section 9 state which convention was used.

The experiments divide cleanly by what a reader must obtain in order to repeat them. The verification and simulator experiments generate their own data from the simulator described in Section 8.3, which is distributed with the accompanying package; they therefore require no download, no dataset licence and no deep-learning framework, and run under a single deterministic entry point that records library versions and digests alongside the results. This is a deliberate property rather than an incidental one: the mathematical claims of Sections 5 and 7 can be checked in full without access to any measured data, and their validity does not rest on either bearing benchmark.

The field experiments necessarily require the CWRU and NASA IMS archives. Both are publicly available, but they are externally maintained and are not redistributed with the package, so reproducing those results involves a separate acquisition step and the training framework used for the shared representation; the corresponding procedure is documented separately. Because the calibration analyses operate on the one-dimensional residual streams alone, they can be repeated from exported residuals without retraining and without redistributing either dataset.

9. Results

Results are reported in the order of Section 8. Throughout, the target coverage is $1 - \alpha = 0.98$, false-alarm rates are out-of-sample, and every quantity produced by the accompanying package is reproducible from the recorded seed.

9.1. Verification of the calibration theory

Exhaustive enumeration. For every combination of $n \in \{4, \dots, 7\}$ and $\alpha \in \{0.2, 0.25, 0.3, 0.4, 0.5\}$ admitted by Corollary 6, enumerating all $(n + 1)!$ orderings reproduces the coverage identity of Proposition 7 exactly, to machine precision, and the rank of the future observation is confirmed equiprobable over $\{1, \dots, n + 1\}$. The feasibility condition and the degenerate-range identity of Corollaries 6 and 7 were checked against their closed forms over a grid of eight values of α spanning 0.005 to 0.25, with agreement at every integer n in a range extending well beyond the boundary in each case. These are checks of the statements themselves rather than of a sampling approximation to them.

The estimator–stopping design. Table 1 reports the four arms across the six residual distributions. Two patterns are visible and neither depends on the distribution.

Table 1

Out-of-sample false-alarm rate (%) by arm and residual distribution, nominal level 2.0%. A1 reproduces the published pipeline; A4 is the configuration to which Proposition 7 applies without qualification. The last column is the median commissioning length selected by the adaptive rule in arm A1.

Residual distribution	A1	A2	A3	A4	med. n (A1)
lognormal(0,0.4)	3.43	1.41	3.74	2.16	96
gamma(k=3)	3.09	1.62	3.55	1.81	97
exponential(1)	2.83	1.26	3.72	1.74	119
pareto(a=3)	3.61	1.28	3.87	1.77	118
halfnormal	3.14	1.41	3.74	1.89	60
bimodal mixture	3.47	1.54	3.79	1.82	66
pooled coverage	0.96737	0.98580	0.96264	0.98136	
(standard error)	(0.00119)	(0.00058)	(0.00043)	(0.00063)	
pooled FAR (%)	3.26	1.42	3.74	1.86	

Table 2

Standard deviation of the certified radius against commissioning length, on a lognormal residual stream. Inside the degenerate range of Corollary 7 the estimator is the sample maximum and its dispersion is essentially constant; it contracts only once an interior order statistic becomes available.

n	k_α	sample max?	sd(τ)
49	49	yes	0.4999
60	60	yes	0.5028
80	80	yes	0.5027
98	98	yes	0.4987
99	98	no	0.3061
120	119	no	0.3081
200	197	no	0.2012
400	393	no	0.1340

First, arm A4 attains a pooled coverage of 0.98136 against the value $k_\alpha/(n+1) = 50/51 = 0.98039$ predicted exactly by Proposition 7, a discrepancy of 1.5 standard errors over 900 trials. The agreement holds for every member of a family chosen to be as dissimilar as the tails, supports and modalities allow, which is the behaviour a distribution-free statement requires and the behaviour whose absence would falsify it.

Second, the two conventions separate cleanly. Decomposing the coverage loss gives -1.87 percentage points attributable to interpolation at fixed n , and -1.84 under adaptive stopping; the effect is stable in sign and magnitude across all six distributions, exactly as Remark 20 predicts. The corresponding entries for the stopping rule are $+0.44$ and $+0.47$ percentage points, that is, in the opposite direction and roughly four times smaller. Interpolation, not adaptive stopping, is the dominant source of the excess false-alarm rate.

Isolating stopping selection from sample size. The comparison just made conflates two effects, because the adaptive rule does not stop at $n = 50$ but at a median of 68 observations, and inside the degenerate range a larger n yields a larger radius. Comparing the adaptive rule against a fixed- n calibration on an independent block of the *same* realized size isolates the selection effect proper: it is -0.11 , $+0.19$, $+0.01$, -0.18 , -0.06 and $+0.00$ percentage points across the six distributions. The magnitude is at most 0.2 points and the sign is not consistent, so the selection mechanism described in Remark 21 is negligible at this configuration and the apparent conservatism of the adaptive arm is a sample-size effect.

Estimator variance and the degenerate range. Table 2 gives the standard deviation of the certified radius as a function of n .

Table 3

Rejection rate of the permutation test at level 0.05. The first column is the size, measured on exchangeable streams; the remainder is the power against an injected monotone drift, in standard deviations of accumulated trend.

Residual distribution	size	drift 0.5	drift 1.0	drift 2.0	drift 3.0
lognormal(0,0.4)	0.042	0.133	0.692	1.000	1.000
gamma(k=3)	0.058	0.233	0.692	1.000	1.000
exponential(1)	0.042	0.408	0.825	1.000	1.000
pareto(a=3)	0.025	0.683	0.992	1.000	1.000
halfnormal	0.083	0.208	0.592	0.992	1.000
bimodal mixture	0.042	0.525	0.875	1.000	1.000

The dispersion is flat to within 1% across [49, 98] and then falls by a factor of 1.6 at the first interior order statistic and by 2.5 by $n = 200$. Clearing the commissioning floor is therefore necessary but not sufficient: a commissioning length chosen just above 49 satisfies Corollary 6 while purchasing the most variable estimator available.

The exchangeability test. Table 3 reports its size and power. The size column lies between 0.025 and 0.083 against a nominal 0.05, within the binomial standard error of 0.0199 at 120 trials, and shows no systematic dependence on the distribution, as a rank-based statistic requires. Power rises monotonically with the injected drift and reaches unity at two standard deviations of accumulated trend in five of six cases. The test is therefore usable as a probe of Assumption 4 rather than as a source of spurious alarms.

9.2. Controlled instantiation on the simulator

The representation operator and its training. The operator is the convolutional autoencoder of the technical note, reproduced without modification at 847,601 trainable parameters and fitted on 4000 simulated nominal windows. The epoch count is a budget rather than a target: training is stopped by validation loss, the learning rate is annealed on plateaus, and the best weights are restored. On the run reported here it terminated after 372 of a permitted 500 epochs, with the best validation loss of 2.62×10^{-3} attained at epoch 347 and the learning rate annealed from 10^{-3} to 2×10^{-6} in nine reductions. Validation uses a held-out split of the *nominal* data only, so early stopping selects for reconstructing nominal behaviour and cannot consult a fault.

One operator, trained once, is used for every experiment in this subsection. Deterministic kernels are enabled throughout.

End-to-end membership. With the radius certified from 120 nominal windows, the held-out false-alarm rate on 1500 further nominal windows is 2.20% against a nominal 2%. The median residual rises from 2.32×10^{-3} on nominal windows to 6.88×10^{-3} , 1.08×10^{-2} and 1.73×10^{-2} on ball, outer-race and inner-race faults, and the single membership rule (25) detects them at 91.5%, 100% and 100% respectively. No fault observation entered the fitting, the calibration or the stopping decision.

Four simulated assets differing only by an installation signature, sharing one representation operator and one target coverage, identify radii of 1.00×10^{-2} , 8.26×10^{-3} , 2.45×10^{-2} and 1.43×10^{-2} , a spread of 2.97 between the largest and the smallest. This is the behaviour Remark 12 predicts: the representation is shared and its residual scale is not.

Intrinsic dimension of the nominal cloud. Table 4 tests the prediction of Remark 4 that each nuisance parameter acting non-trivially contributes one direction to the nominal set.

Each additional parameter raises the estimate by close to one, and the single-parameter case is recovered exactly. The residual gap grows with the dimension, which is the known negative bias of the estimator rather than an unexplained discrepancy; a mistaken reading would fail to track the count at all. Figure 2 shows the tracking directly.

Additive noise thickens the cloud without adding dimension, and the estimator shows this as scale dependence: on the full simulator the estimate falls from about twelve at the smallest neighbourhood to close to four at the largest, approaching the noise-free values. Against an ambient dimension of 1200, an intrinsic dimension near four is what makes the covering argument of Remark 16 usable rather than vacuous, and it is measured here rather than assumed.

Table 4

Maximum-likelihood intrinsic dimension of the noise-free nominal family, against the number of active nuisance parameters. The estimator is biased downward, increasingly so with dimension, so the criterion is whether it *tracks* the prediction rather than whether it matches it. Ambient dimension is 1200.

active nuisance parameters	predicted	$k = 10$	$k = 20$	$k = 40$
amplitude	1	1.00	1.00	0.99
amplitude+phase	2	1.97	1.95	1.94
amplitude+phase+speed	3	2.88	2.85	2.80
all four	4	3.68	3.59	3.48
phase+speed+offset	3	2.84	2.76	2.70
speed+offset	2	1.93	1.92	1.91

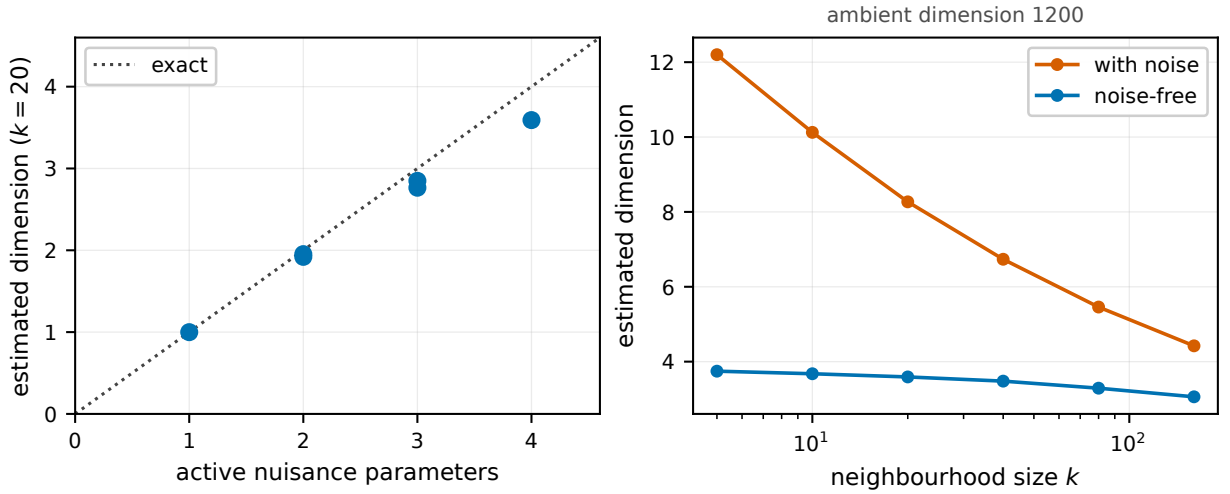


Figure 2: Intrinsic dimension of the nominal cloud. Left: the estimate against the number of nuisance parameters actually varied, with the diagonal marking exact recovery; the estimator tracks the count with a downward bias that grows with dimension. Right: scale dependence on the full simulator, where additive noise raises the estimate at small neighbourhoods and the manifold structure emerges as the neighbourhood grows. Generated by `experiments/make_figures.py`.

Attractor drift under slow parameter change. On chronological records of 700 windows whose first half is nominal and whose severity then ramps smoothly, the radius is commissioned on the early-life segment, frozen, and never adapted. Departure is declared by a persistence rule. Onset occurs at record 351; persistent departure is declared at 473, 496 and 545 for inner-race, outer-race and ball faults, delays of 122, 145 and 194 records, at severities of 0.211, 0.251 and 0.335 on a ramp reaching 0.6. Post-departure exceedance is 96.1%, 98.2% and 96.8%. Figure 3 shows the trajectories.

The false-alarm rate measured after commissioning but before onset, where the process is still nominal, is 5.2% against the 2% target. This is higher than the calibrated level and is worth stating: it is measured on a single realisation over roughly 230 records, so its sampling error is substantial, but it also reflects that the commissioning segment and the later nominal segment are not perfectly exchangeable in a record that will subsequently degrade. It is the same assumption, Assumption 4, that the permutation test of Section 9.1 is designed to probe.

What the detector is given is no severity information, no fault label and no adaptation after commissioning. What it reports is failure of membership in a region estimated before any degradation occurred, which is the distinction drawn in Section 10.5 between motion along a fixed orbit and motion of the orbit itself.

Saturation of the nominal radius. Table 5 measures the radius of Definition 3 against a reference estimated from 6000 held-out nominal windows. Attained coverage is measured on a fresh evaluation block of 300 windows drawn,

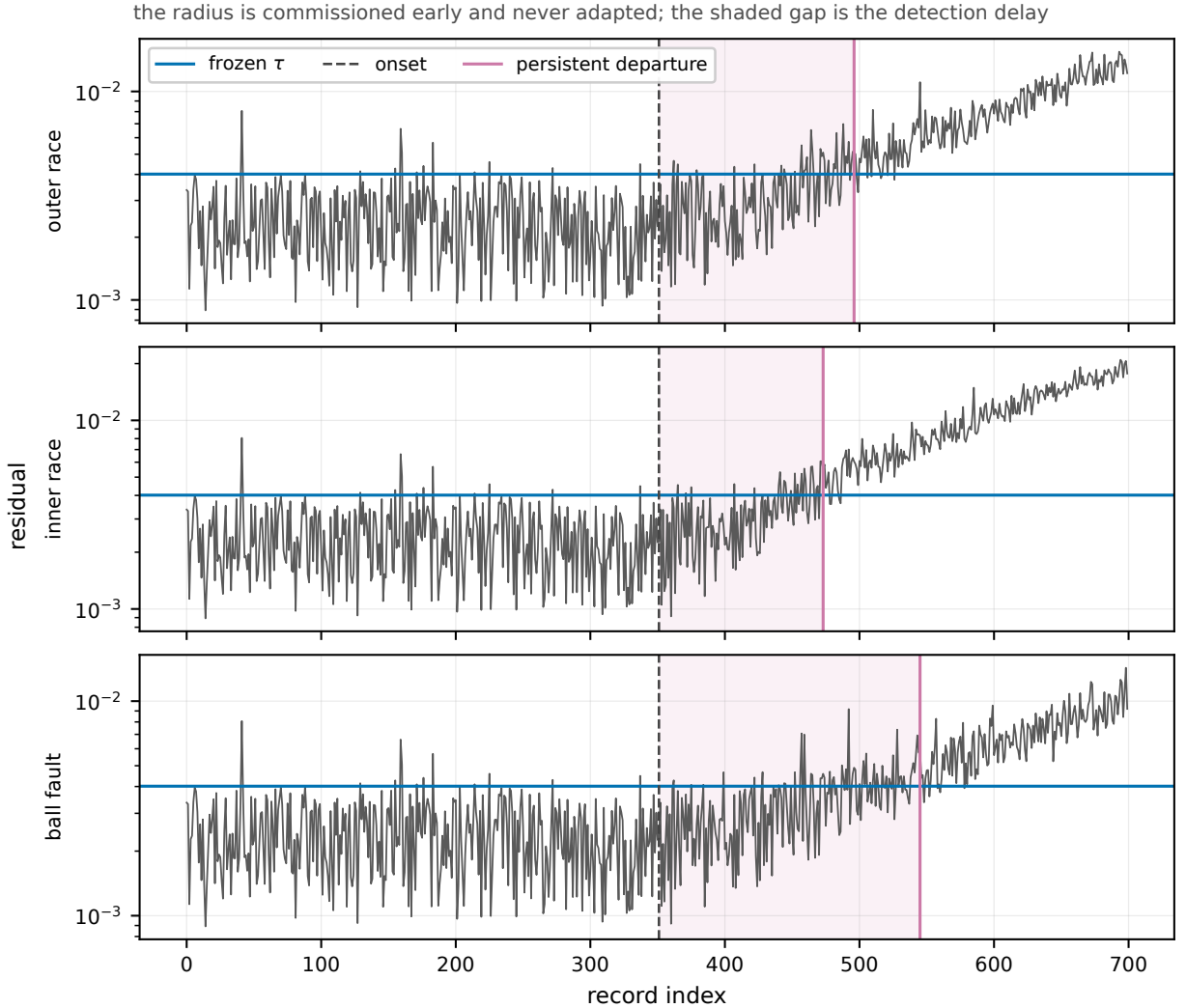


Figure 3: Degradation trajectories under a frozen radius. The horizontal line is the radius commissioned on the early-life segment and never afterwards adapted; the dashed line marks the onset of non-zero severity and the solid vertical line the first persistent departure, with the shaded interval the detection delay. Generated by `experiments/make_figures.py`.

disjointly from the calibration draw, on each of the 200 repetitions. The choice is not a detail. Holding a single evaluation block fixed estimates coverage against that block's empirical distribution function rather than against the distribution, an offset of order $\sqrt{p(1-p)/M}$ that does not average away over repetitions and is of the same order as the excess coverage being reported. Drawing calibration and evaluation points jointly without replacement from one pool is moreover not an approximation to independent sampling: the pool is exchangeable, so any $n+1$ of its members drawn without replacement are exchangeable, which is the sole hypothesis of Proposition 7.

Four things follow, and the first three are the substance of Remark 8.

First, the certified coverage agrees with the exact value $k_a/(n+1)$ at every commissioning length on the grid, to within 1.1 Monte Carlo standard errors. This is a sharper statement than the inequality coverage $\geq p$, and it is the sharper statement that is worth making: $k_a/(n+1)$ is not monotone in n and takes distinctly non-trivial values inside the degenerate range, where it rises to 0.9899 at $n = 98$ against an observed 0.9889. An estimator that merely covered conservatively would satisfy the inequality while missing that structure entirely.

Table 5

Saturation of the nominal radius against the reference $R_p = 22.599$. “certified” is the estimator of Corollary 4, “interp” the interpolated empirical quantile. “exact” is the coverage predicted by Proposition 7, $k_n/(n+1)$; the observed certified coverage matches it to within 1.1 Monte Carlo standard errors at every n , including the non-trivial values taken inside the degenerate range. The interpolated estimator undercovers throughout. A dash marks a commissioning length below the floor, at which no radius is certifiable.

n	certified	sd	bias (%)	coverage	exact	interp	int. cov	max?
30	—	—	—	—	—	21.109	0.9499	—
49	23.106	1.606	+2.24	0.9801	0.9800	21.570	0.9611	yes
60	23.613	1.726	+4.49	0.9841	0.9836	21.708	0.9655	yes
80	23.860	1.522	+5.58	0.9868	0.9877	21.974	0.9683	yes
98	24.110	1.534	+6.69	0.9889	0.9899	22.019	0.9706	yes
99	22.917	1.300	+1.41	0.9805	0.9800	22.084	0.9704	no
150	22.812	0.993	+0.94	0.9805	0.9801	22.216	0.9736	no
300	22.671	0.723	+0.32	0.9795	0.9801	22.388	0.9763	no
800	22.619	0.407	+0.09	0.9799	0.9800	22.518	0.9786	no
2000	22.584	0.225	−0.07	0.9794	0.9800	22.538	0.9789	no

Second, inside the degenerate range the certified radius does not settle: its mean drifts from 2.24% above R_p at $n = 49$ to 6.69% at $n = 98$, and then falls discontinuously to 1.41% at $n = 99$, where an interior order statistic first becomes available. This is not a property of the regime but of the estimator, which is the sample maximum throughout that band by Corollary 7. A saturation curve computed this way cannot flatten there, and apparent settling is the maximum of a growing sample. The excess coverage inside the band is the same fact stated in the other coordinate: covering more than the target is precisely what a radius that is too large does.

Third, the interpolated estimator behaves in the complementary way. It approaches R_p smoothly from below, producing the familiar saturating curve, and its coverage is below the $p = 0.98$ target at every commissioning length examined, from 0.950 at $n = 30$ to 0.979 at $n = 2000$. The visual impression of convergence and the coverage guarantee are supplied by different estimators, and not by the same one.

Fourth, replacing the split-sample reference by a plug-in reference fitted on the same observations shrinks the radius by 2.2 to 2.7% at $n \in \{49, 98, 300, 800\}$, confirming the sign predicted in Remark 6. Coverage, however, is not degraded, because the evaluation distances are recomputed in the same plug-in geometry and the two shift together. The practical cost of a plug-in reference is therefore the comparability of radii across assets and regimes, not the validity of the membership test, and the remark is refined accordingly.

A negative result: the radius is not a detector. Measured under the shared nominal projection, the certified radius of each regime is 23.13 for nominal, 22.63 for outer-race, 22.78 for inner-race and 23.95 for ball faults, that is, ratios to nominal between 0.98 and 1.04. There is no separation. The explanation is mechanical: a two-dimensional principal projection of a raw vibration window is dominated by the shaft harmonics, while the fault signature is a low-amplitude impulse train contributing almost nothing to the leading variance directions. The same simulator, the same nominal-only fitting and the same membership rule detect those faults at 91.5–100.0% when the residual is used instead. Saturation of a region and separability of that region from its neighbours are therefore independent properties, as Remark 9 states, and a saturation plot is evidence about the sampling of the nominal geometry rather than about regime discrimination.

Operator adequacy: coverage cannot choose between residuals. Table 6 applies the probe of Remark 15 to two operators calibrated identically on the same nominal data, at the same target coverage.

The linear operator accepts a point at 1160 units from the nearest nominal window, some 380 times the nominal spacing, while assigning it a residual of order 10^{-28} . Its accepted region is a slab around the retained subspace, not a tube around the nominal manifold, exactly as Remark 15 anticipates for a residual that measures distance to a structure containing the nominal set rather than to the set. The convolutional operator rejects everything beyond 1.6 times the spacing, and its residual rises from 1.08×10^{-3} to 38.6 across the probe, a growth of four orders of magnitude.

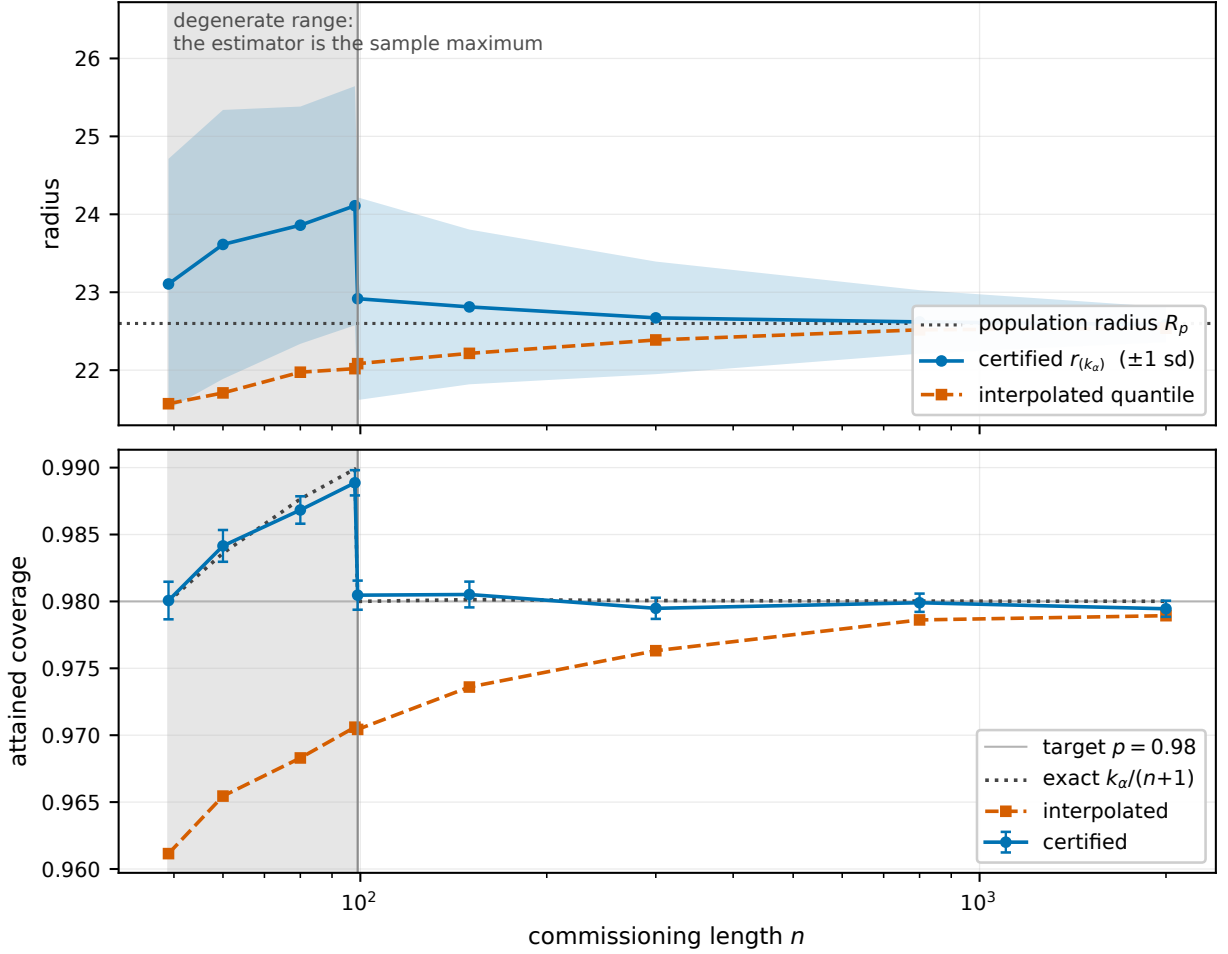


Figure 4: Saturation of the nominal radius. Upper panel: the certified radius with a ± 1 standard-deviation band, against the population value; the shaded region is the degenerate range of Corollary 7, in which the estimator is the sample maximum and therefore rises with n rather than settling, and the vertical rule marks the first commissioning length admitting an interior order statistic. Lower panel: attained coverage, with Monte Carlo error bars. The dotted reference is not the target level but the exact coverage $k_a/(n+1)$ of Proposition 7, which exceeds the target and is not monotone in n ; the certified curve tracks it, while the interpolated estimator produces the smooth radius curve above and undercovers at every length examined. Generated by `experiments/make_figures.py`.

The comparison is informative precisely because detection does not separate the two. Both reach 100% on outer- and inner-race faults, and the linear operator reaches 84% on ball faults against 90% for the convolutional one. A reader judging by detection rates alone would find them comparable. Coverage does not separate them either, by Proposition 7, which holds for both. Only the probe does, and the property it detects is the one on which tightness depends. This is the empirical content of the claim that this work certifies the level and inherits the shape.

9.3. Field transfer: CWRU

The results in this subsection and the next are carried over from the existing pipeline and are not regenerated by the accompanying package, for the reasons given in Section 8.7. By Section 8.6 they were obtained with the interpolated estimator and $N_{\min} = 40$, that is, in the A1 configuration, and therefore do not carry the guarantee of Proposition 7.

Table 7 establishes the reference. Independently trained asset-specific detectors, one representation and one preprocessing transformation per operating condition fitted on that condition's complete healthy record, achieve 100%

Table 6

Distance-proxy probe. Latent codes are pushed away from the nominal range, decoded, and the residual compared against the true distance to a disjoint nominal cloud, whose median nearest-neighbour spacing is 3.04. Both operators attain the calibrated coverage; they differ by more than two orders of magnitude in what they accept.

operator	healthy FAR (%)	ball DR (%)	max accepted distance	ratio to spacing	adequacy
PCA-16	2.0	84	$1.16e+03$	381.6×	fail
Conv1D AE	2.3	90	4.98	1.6×	pass

Table 7

Asset-specific reference on CWRU. Each detector is trained and calibrated independently on the complete healthy record of its operating condition. This reference is deliberately favourable and operationally unavailable.

RPM	threshold	healthy FAR (%)	ball DR (%)	inner DR (%)	outer DR (%)
1730	0.6485	2.23	100.00	100.00	100.00
1750	0.7590	2.23	100.00	100.00	100.00
1772	0.7248	2.23	100.00	100.00	100.00
1797	0.4591	2.46	100.00	100.00	100.00
mean	–	2.29 ± 0.12	100.00	100.00	100.00

Table 8

Simulation-to-field transfer on CWRU under one shared representation. Only the information available for estimating the radius differs between strategies. Healthy FAR covers the complete healthy sequence including commissioning observations; post-convergence values appear in Table 9.

RPM	calibration	τ	FAR (%)	ball (%)	inner (%)	outer (%)	windows
1730	direct transfer	0.1027	100.00	100.00	100.00	100.00	–
1730	field oracle	0.3578	2.23	99.75	100.00	99.86	–
1730	online	0.3520	3.71	99.75	100.00	100.00	49
1750	direct transfer	0.1027	100.00	100.00	100.00	100.00	–
1750	field oracle	0.3434	2.23	100.00	100.00	100.00	–
1750	online	0.3440	1.98	100.00	100.00	100.00	62
1772	direct transfer	0.1027	100.00	100.00	100.00	100.00	–
1772	field oracle	0.3512	2.23	100.00	100.00	100.00	–
1772	online	0.3550	1.24	100.00	100.00	100.00	58
1797	direct transfer	0.1027	100.00	100.00	100.00	100.00	–
1797	field oracle	0.3886	2.46	98.03	100.00	100.00	–
1797	online	0.3869	3.45	98.52	100.00	100.00	49

detection for all three fault families at a mean healthy false-alarm rate of $2.29 \pm 0.12\%$. Their absolute decision thresholds range from 0.459 to 0.759, a spread of 1.65, although their diagnostic operating points are identical. Numerical residual scales are not interchangeable even between models of the same architecture trained on the same data, which is Remark 12 in its sharpest form.

Table 8 reports the three calibration strategies under one shared simulator-trained representation.

Direct transfer of the absolute simulator threshold fails under every operating condition, flagging every healthy window as anomalous while producing an apparent 100% detection rate. This is the predicted outcome of Remark 12 and is the empirical content of the claim that the representation transfers and its scale does not: the same representation, given a locally estimated radius, recovers an operationally meaningful boundary in the same table.

Table 9

Agreement between the converged online radius and the complete-data oracle. Post-calibration FAR is evaluated only on healthy observations occurring after the declared convergence point.

RPM	τ_{oracle}	τ_{online}	rel. diff. (%)	windows	post-cal. FAR (%)
1730	0.3578	0.3520	1.64	49	3.94
1750	0.3434	0.3440	0.19	62	1.75
1772	0.3512	0.3550	1.06	58	0.87
1797	0.3886	0.3869	0.44	49	3.90
mean	—	—	0.83	54.5 ± 6.6	2.62

Local calibration does exactly that. The complete-data field oracle reproduces the expected 2% operating point, and the online procedure reaches a comparable one after 49, 62, 58 and 49 observations, a mean of 54.5 ± 6.6 . Table 9 compares the converged radii with the oracle values.

The converged radii differ from their oracle references by 0.19–1.64%, a mean of 0.83%. The procedure therefore recovers nearly the same acceptance boundary that retrospective access to the complete healthy record would have produced, from at most 62 observations.

The commissioning lengths, and what they do not show. The four commissioning lengths are 49, 62, 58 and 49, and the coverage floor of Corollary 6 at $\alpha = 0.02$ is also 49. By Remark 25 these coincide for unrelated reasons: the stopping rule cannot emit a length below 49 at the configuration used, and that bound does not involve α . The clustering is therefore uninformative about the floor, and is reported here only to forestall the inference.

More consequentially, all four lengths lie in [49, 98], which by Corollary 7 is the range in which a certified radius would be the sample maximum. Combined with the use of an interpolated quantile, this places the field calibration in precisely the regime that Table 5 characterizes as simultaneously the most variable and the least conservative.

Robustness, read as a probe of exchangeability. Permuting the commissioning observations and their arrival order over five realizations per condition, with the representation and all parameters fixed, gives 20 convergent runs out of 20. The estimated radius stays within $2.08 \pm 0.71\%$ of the oracle and the fleet-averaged post-calibration false-alarm rate is $3.59 \pm 1.59\%$. Following Remark 22, the absence of systematic dependence on arrival order is the evidence bearing on Assumption 4, and it is consistent with the assumption holding within the commissioning window.

The dispersion is nonetheless substantial: one 1750 RPM realization produced a post-calibration false-alarm rate of 13.8% while remaining within 4% of the oracle radius. This is finite-sample uncertainty in an extreme quantile estimated inside the degenerate range, and Table 2 gives its scale directly: at $n \approx 55$ the estimator has the same dispersion as at $n = 49$, and 2.5 times that available at $n = 200$.

Cost of sharing one representation. Relative to the asset-specific reference of Table 7, replacing four independently trained detectors by one simulator-trained representation with local calibration changes mean ball-fault detection from $100.00 \pm 0.00\%$ to $99.57 \pm 0.71\%$, leaves inner- and outer-race detection at 100%, and raises the mean healthy false-alarm rate from $2.29 \pm 0.12\%$ to $2.60 \pm 1.18\%$. The reference requires a fitted preprocessing transformation, a trained representation and the complete healthy record of every condition; the shared alternative requires 49–62 nominal observations and no field training.

9.4. Field transfer: NASA IMS

Table 10 reports the four run-to-failure bearings.

Calibration converged for all four bearings after 49–57 recordings. The converged radii span 6.67×10^{-3} to 1.89×10^{-2} , a factor of 2.8, under an identical representation, an identical target coverage and an identical stopping rule. This is the same phenomenon as the simulated assets of Section 9.2 and as the CWRU threshold spread of Table 7, now on physically distinct units: the transferable object is the representation, and the radius is individual. Corollary 2 is the structural statement behind it, applied per physical unit rather than per operating condition.

All four bearings subsequently departed persistently from their frozen neighbourhood, with a mean post-departure exceedance of 94.61% and lead times of 14.5 to 74.0 hours, a mean of 45.4. Repeating the commissioning with five permuted early-life samples per bearing gave 20 convergent runs out of 20, radii within 0.83% of their references on

Table 10

NASA IMS deployment under one shared simulator-trained representation with local radius estimation. Relative error is against the complete-data reference for that bearing; lead time is measured to the documented end of the experiment.

Bearing	windows	τ_k	rel. error (%)	post-dep. exceedance (%)	lead time (h)
1	49	0.010680	0.10	99.12	74.00
2	57	0.014950	0.35	86.42	38.83
3	49	0.018930	2.87	95.88	14.50
4	49	0.006672	0.18	97.02	54.33
mean	51	–	0.87	94.61	45.42

average, and a healthy false-alarm rate of 4.98%; bearing 3, which shows the largest deviation in Table 10, is also the most sensitive to commissioning-sample variability, at $1.58 \pm 1.88\%$.

Two caveats attach to the lead times specifically. They depend on the persistence rule and on the recording-level aggregation, and Section 8.7 records that the source code and the accompanying text disagree on both: three exceedances in five recordings against eight in ten, and a mean against a 95th-percentile aggregation. The values above are those produced by the code as executed. Resolving the discrepancy is a prerequisite for quoting lead times as a headline result, and is listed in Section 12.

9.5. Summary

The verification experiments confirm Proposition 7 exactly by enumeration and to within 1.5 standard errors by simulation across six dissimilar residual distributions, and they identify interpolation rather than adaptive stopping as the dominant source of the excess false-alarm rate observed in practice. The simulator experiments exercise the full chain with the operator of the technical note and supply four results that qualify how the framework should be read: the intrinsic dimension of the nominal cloud tracks the number of active nuisance parameters, which is what makes the covering argument usable at ambient dimension 1200; the certified radius cannot flatten inside the degenerate range; the radius does not discriminate between regimes that the residual separates almost perfectly; and two operators with comparable detection rates and identical calibrated coverage differ by a factor of 240 in what their accepted regions admit, so neither detection nor coverage suffices to choose between them. The field experiments show that the representation transfers while its numerical scale does not, that local calibration recovers the complete-data operating point from at most 62 observations, and that independently commissioned physical units identify radii differing by a factor of 2.8 under one shared representation. The field calibration, however, was performed with the estimator and at the commissioning lengths that Sections 7.4 and 8.6 identify as uncertified and maximally variable; re-running it is the single most valuable extension of this work.

10. Discussion

10.1. What the quotient construction delivers, and what it does not

The results are best read as an asymmetry. Four things follow from the geometry, and one conspicuously does not.

Diagnosis factors through the quotient (Proposition 2), so a correct diagnostic decision exists as a function of the equivalence class alone; this is what licenses seeking a representation at all rather than a per-asset classifier. The nominal region is compact and hence has a finite radius (Proposition 1, Corollary 3), so there is a well-defined finite quantity to estimate, which is not automatic and does not follow from the informal saturation criterion on its own. That radius admits a distribution-free estimate from a computable number of observations (Proposition 7, Corollary 6), which converts an engineering threshold into a certified quantity. And regions once recovered can be labelled and calibrated individually with per-class validity (Corollary 5), so the taxonomy can grow after deployment without recalibrating what already exists.

What does not follow is separation. Nothing in the construction bounds the probability of *admitting* a competing region, and Remark 10 shows the guarantee is fully compatible with a permanently ambiguous diagnosis. The framework is therefore a theory of detection with a placeholder where classification should be, and the placeholder is structural rather than an artefact of the particular operator: a scalar residual is many-to-one and discards the direction of departure, and the empirical result of Section 9.2 shows that the low-dimensional radius which describes the nominal

geometry does not discriminate the regimes it describes. Stating this plainly seems preferable to presenting an $\arg \min$ over regions as though it inherited the guarantee that the membership test does.

10.2. The transferable object is the representation, not its scale

The claim that a representation transfers while its residual scale does not now rests on three independent observations of increasing force.

Simulated assets differing only by an installation signature identify radii spanning a factor of 2.49 under one operator. Physically distinct NASA IMS bearings, independently commissioned under one representation and one convergence criterion, identify radii spanning a factor of 2.8. The sharpest case, however, is the CWRU reference of Table 7: four detectors of identical architecture, trained on the data of their own operating condition, produce decision thresholds from 0.459 to 0.759 while achieving indistinguishable diagnostic operating points. Here the variation cannot be attributed to the asset, the sensor or the operating point, because it survives when those are held fixed and only the optimization trajectory differs.

The conclusion is stronger than a deployment convenience. Diagnostic performance is a property of the learned geometry; the numerical value of the residual is a property of a particular fit. Transferring the former and re-estimating the latter is therefore not a heuristic for saving effort but the only combination consistent with what the residual actually is, and Remark 12 gives the reason: the membership predicate inherits orbit-invariance from Proposition 2, while the score used to evaluate it does not.

10.3. Saturation, re-read

Three findings revise how the saturation evidence of the companion work [9] should be interpreted, and the revisions follow from proofs rather than from anything discovered in the data.

The first concerns the estimator. Between $1/\alpha - 1$ and $2/\alpha - 1$ the certified radius is the sample maximum (Corollary 7), which increases with n ; the measured drift is from +2.13% to +6.43% of R_p across that band, followed by a discontinuous fall. A curve computed this way cannot flatten inside the band, so flattening observed there is not evidence that the geometry has stabilized.

The second concerns which estimator produces the familiar picture. The interpolated quantile approaches R_p smoothly from below and yields the reassuring saturating curve, and it undercovers at every commissioning length examined. The visual and the guarantee are supplied by different estimators. This is not a reason to abandon saturation plots, but it is a reason to report attained coverage beside them, since the plot alone cannot distinguish a saturated regime from a conveniently biased statistic.

The third is the negative result. Fault regimes measured under the shared nominal projection have radii within 4% of nominal, while the residual detects the same faults at 79–100%. Saturation characterizes how well a region has been sampled; it says nothing about whether that region is distinguishable from its neighbours. Evidence of the first has occasionally been offered in support of the second, including in the framing that motivated this work, and the two should be kept apart.

10.4. Relation to conformal prediction

The calibration machinery is a distribution-free tolerance limit in the sense of Wilks [29], and the multi-region rule (18) is the class-conditional form of a conformal predictor [24, 28]. It is worth being explicit that this was arrived at from the quotient geometry rather than imported: the radius is what the compactness result says exists, and exchangeability is what the nominal regime supplies during commissioning.

The connection is useful in both directions. It supplies the guarantee at essentially no cost, since the estimator is an order statistic of quantities the pipeline already computes. It also identifies precisely where the present framework is weak, because the gap named in Remark 10 is the distinction between validity and efficiency that conformal prediction has studied at length: a predictor can be valid and useless if its prediction sets are large. Read this way, the classification problem left open in Section 6 is not an unexplored question but a known one, and the literature on conditional validity and on efficient non-conformity scores is the natural place to look for a construction that tightens $\Gamma(x)$ without sacrificing Corollary 5.

10.5. Two dynamical readings, and why they must be kept apart

The embedding reading of Remark 1 invites a second, superficially similar statement about the run-to-failure trajectories, and the two are worth separating because conflating them would misdescribe what the NASA experiment shows.

The first is reconstruction at fixed parameters. For a bearing in a stable operating regime, the delay-coordinate window reconstructs an invariant set whose geometry the nuisance group acts on in the manner of Remark 4. This is the statement that justifies the window as the observation object and that predicts the intrinsic dimension of the nominal cloud. It concerns a single regime and says nothing about change over time.

The second is drift of the invariant set itself. Degradation is not motion along a fixed attractor but slow variation of the parameters that define it, so the reconstructed set deforms over the life of the unit while the commissioning radius stays frozen where it was estimated. The persistent departures observed on all four IMS bearings are departures of this kind: the region has moved, not the point within it. This is a bifurcation-theoretic picture rather than an embedding-theoretic one, and Takens' theorem has nothing to say about it.

Keeping them apart also clarifies what the frozen radius is doing. It is a boundary estimated for one configuration of the system and then held fixed while the system is allowed to change; the alarm is informative precisely because the boundary is *not* adapted. An adaptive radius would track the drift and absorb the degradation into the baseline, which is the failure mode that motivated confining adaptation to a finite commissioning phase in Section 7.5.

10.6. Interpretability as a structural property

The decision produced by (25) is a statement that an observation lies outside a region estimated from that asset's own nominal behaviour, with a stated coverage. Nothing about it requires a post-hoc attribution method to be understood: the quantity, the reference set and the level are all part of the construction, and the reported number is an order statistic of observed residuals rather than an internal activation.

This is the distinction argued for by Rudin [16] in a different setting, and it applies with some force here because the diagnostic decision is consequential and the audit question is concrete. Asked why an asset was flagged, the framework answers that its residual exceeded a radius estimated from a specified number of its own nominal observations at a stated coverage; asked how confident that boundary is, it answers with Proposition 7 and, where the commissioning length falls inside the degenerate range of Corollary 7, with the caveat that the estimate is the sample maximum.

Two qualifications keep the claim honest. The representation operator remains opaque: nothing here explains *why* a particular window has a large residual, only that it does and that this is anomalous relative to a calibrated region. And interpretability at the level of the decision rule does not confer it at the level of fault type, which Section 6 leaves open. What the construction provides is an auditable membership criterion, not an explanation of the underlying mechanics.

10.7. Consequences for practice

Five recommendations follow, each from a proved statement rather than from tuning.

1. Calibrate with the order statistic $r_{(k_\alpha)}$ rather than with an interpolated empirical quantile. The interpolated value lies at or below it and does not satisfy the coverage bound; the measured cost is 1.9 percentage points of coverage, consistently in sign across every distribution examined.
2. Require $n \geq 2/\alpha - 1$, not merely $n \geq 1/\alpha - 1$. Clearing the floor makes a radius certifiable; clearing the degenerate range makes it stable, and the difference is a factor of 2.5 in the estimator's dispersion.
3. Report attained coverage alongside any saturation curve, for the reason given in Section 10.3.
4. Treat permutation of commissioning order as a test of Assumption 4 rather than as a robustness check. The test has nominal size and high power against drift, so a systematic dependence on arrival order would be informative rather than merely inconvenient.
5. Estimate the reference point, and any projection, on observations disjoint from those used to compute the radius. The measured plug-in shrinkage is 2–3%; coverage survives, but radii computed in different plug-in geometries are not comparable across assets, which is what a fleet-level view requires.

None of these requires additional data, additional labels, or a different architecture.

10.8. A quantitative reading of the governing thesis

The premise stated in Section 1 is that physically generated signals do not exhibit unbounded or adversarial variation, and that departure from a compact nominal region is what an anomaly is. The results give that premise two quantitative renderings, one structural and one operational, and both are falsifiable.

The structural rendering is Corollary 3: under bounded nuisance the nominal orbit is relatively compact, so the radius is finite and there is something to estimate. Were nominal variation genuinely unbounded, no finite radius would exist and the entire construction would be vacuous rather than merely inaccurate.

The operational rendering is the commissioning floor. At a target coverage of 98%, 49 nominal observations suffice to certify a boundary distribution-free, and 99 suffice to do so with a stable estimator. That a regime can be bounded from tens of observations is the concrete content of the claim that physical variability is compact; a process without that structure would require the radius to keep growing with n , which is exactly what Table 5 shows the estimator doing inside the degenerate range and not doing outside it.

The NASA IMS trajectories close the loop on the diagnostic side. Four independently commissioned bearings, with no fault labels at any stage, departed persistently from their own frozen neighbourhoods with lead times of 14.5 to 74.0 hours. The anomaly signal is not a learned fault signature but the failure of membership in a region whose extent was estimated from roughly fifty nominal observations, which is the inverse-problem claim of this paper reduced to its operational minimum.

11. Limitations

12. Future Work

13. Conclusion

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